

Received 17 December 2025, accepted 10 January 2026, date of publication 14 January 2026, date of current version 28 January 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3654393

 SURVEY

Computer Vision-Based Medical Imaging Techniques: Past, Present, and Future

ALEKSANDR LONCHAKOV^{1,2}, ALEKSANDR SINITCA^{1,3}, (Member, IEEE),
AND DMITRII KAPLUN^{1,4}, (Member, IEEE)

¹Research Center of the Artificial Intelligence Institute, Innopolis University, 420500 Innopolis, Russia

²Department of Automation and Control Processes, Saint Petersburg Electrotechnical University "LETI," 197022 Saint Petersburg, Russia

³Intelligent Devices Institute, Saint Petersburg Electrotechnical University "LETI," 197022 Saint Petersburg, Russia

⁴School of Computer Science and Technology/School of Artificial Intelligence, China University of Mining and Technology, Xuzhou 221116, China

Corresponding author: Dmitrii Kaplun (dikaplun@etu.ru)

This work was supported by the Ministry of Economic Development of the Russian Federation under Agreement 139-10-2025-034 dd. 19.06.2025, IGK 000000C313925P4D0002.

ABSTRACT Medical imaging is fundamental for modern medicine, providing non-invasive techniques for visualizing structures and pathologies that are used for diagnosis, treatment planning, and monitoring of diseases. The year 2025 marks the 130th anniversary of Roentgen's X-ray discovery, which was the inception point of modern medical visualization. This paper examines the historical development, core methodologies, and emerging trends in medical imaging modalities. A detailed examination of key imaging modalities including X-ray, Computed Tomography, Magnetic Resonance Imaging, Ultrasound, Positron Emission Tomography, and Single Photon Emission Computed Tomography is provided, highlighting their features, clinical applications, inherent limitations, and future development directions. Subsequently, the paper provides a comprehensive survey of computer vision-based methods adapted for medical image analysis. It begins by examining the evolution of foundational computer vision models, followed by a discussion of convolutional neural networks (CNNs) and Vision Transformers (ViTs) for image classification tasks. U-Net architectures and transformer-based approaches are examined in relation to image segmentation. In conclusion, current state-of-the-art deep learning models are summarized and potential directions for future research are outlined.

INDEX TERMS Medical imaging modalities, deep learning, classification, segmentation, survey.

I. INTRODUCTION

The crucial role of accurate investigation and diagnosis in the management of diseases management is indisputable. Medical imaging plays a critical role not only in diagnosing diseases, but also in planning and monitoring interventions. Historically, X-ray imaging dominated the field from the late 1890s until the early 1940s [1], serving as the only feasible method to obtain images of internal structures. In contrast, contemporary medical practice now employs a variety of imaging modalities, each capable of acquiring anatomical, physiological, metabolic, and functional information from the human body. Among the most widely used modalities are X-ray, computed tomography (CT), magnetic resonance imaging (MRI), single photon emission

computed tomography (SPECT), positron emission tomography (PET), and ultrasound (US). Each modality offers distinct advantages and is primarily used for a specific range of diagnostic applications. The development and application of these imaging techniques require a complete understanding of both physical principles and biological considerations. To provide a holistic understanding, it is essential to explore not only the basic principles and typical clinical applications of each imaging modality, but also their advantages and inherent limitations. This analysis is critical to guide the appropriate clinical decision making. Furthermore, the field of medical imaging continues to evolve, with significant advances in hybrid imaging systems, which combine modalities such as PET/CT, SPECT/CT, and PET/MRI to provide complementary information for better diagnosis and treatment planning.

The associate editor coordinating the review of this manuscript and approving it for publication was Vishal Srivastava.

Several excellent review papers have already been published that summarize medical imaging modalities, such as those by Seibert [2], Hussain et al. [3], and Wallyn et al. [4], as well as those that cover medical image analysis by Litjens et. al [5], Suzuki [6], and Chen et al. [7]. We extend this body of work by not only summarizing the up-to-date developments of medical imaging modalities that are essential for practitioners but also by providing a discussion of the evolution and current advances in image processing. Furthermore, we emphasize the integration of hybrid imaging technologies and the importance of publicly available datasets in advancing research and clinical applications. This approach aims to bridge the gap between image acquisition and analysis techniques, offering a unified perspective for researchers.

The key contributions of our paper include:

- An up-to-date overview of the history, principles, clinical applications, and limitations of major medical imaging modalities, emphasizing their specific contributions to diagnostic radiology.
- In-depth analysis of the evolution of deep learning is provided in the context of medical image analysis. An emphasis is placed on the adaptation of these techniques from general computer vision applications to medical tasks such as segmentation and classification.
- A comprehensive review of clinically significant medical image analysis tasks and associated datasets is presented, including tumor and organ segmentation, glioma and maxillofacial structure segmentation, cardiac and ophthalmic imaging, as well as microscopic imaging, with insights into the methodologies and performance metrics involved.
- Persistent challenges for medical imaging are discussed, including the need for large annotated datasets. Future research directions are proposed.

The paper is organized as follows. Section III provides an overview of medical imaging modalities, including X-ray, CT, MRI, nuclear imaging, ultrasound, and hybrid systems, detailing their historical evolution. Section IV reviews machine learning methods for medical image analysis, encompassing foundational architectures for classification (e.g., CNNs, Vision Transformers) and segmentation (e.g., U-Net variants), followed by a summary of state-of-the-art (SOTA) applications across various anatomical regions and imaging modalities. Section V synthesizes key insights, discusses unresolved challenges, and outlines future directions for both medical imaging and machine learning applications.

II. MATERIALS AND METHODS

A comprehensive literature search was conducted covering the history of medical imaging, modalities, and deep learning applications. Search terms for medical history and modalities included “history”, “evolution”, “development”, “principles”, “clinical application”, “limitations”, “X-ray”, “radiography”, “computed tomography (CT)”, “magnetic

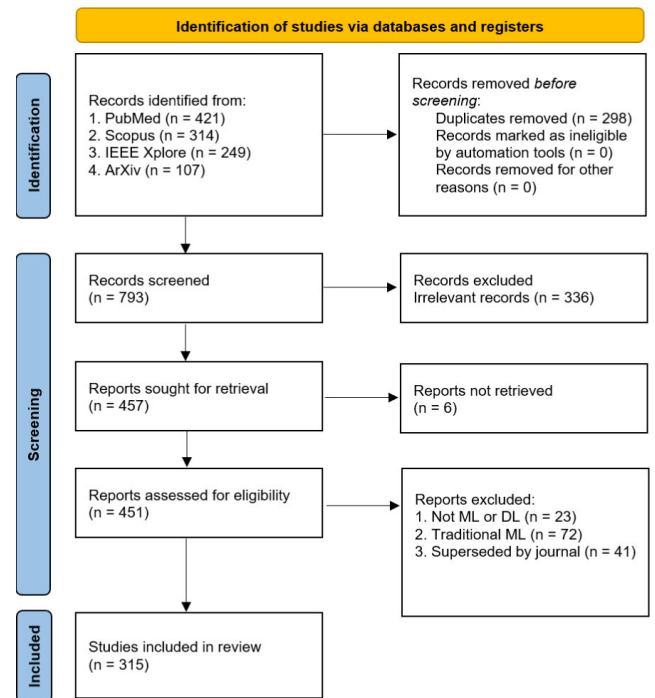


FIGURE 1. PRISMA flowchart summarizing literature search and selection.

resonance imaging (MRI)”, “ultrasound”, “nuclear medicine”, “positron emission tomography (PET)”, “single-photon emission computed tomography (SPECT)”, and hybrid modalities such as “PET/CT”, “SPECT/CT” and “PET/MR”. Keywords list for deep learning and medical applications included “Convolutional Neural Network (CNN)”, “2D CNN”, “3D CNN”, “ResNet”, “DenseNet”, “EfficientNet”, “Inception”, “transformer”, “attention”, “U-Net”, and “Swin-UNETR”, together with task-level terms such as “segmentation”, “semantic segmentation”, “organ segmentation”, “lesion segmentation”, “tumor segmentation”, “classification”, “boundary detection”, “contour extraction”, “detection”, and “localization”. To ensure relevance and exclude general computer-vision papers, search queries employed Boolean operators. The search was performed using electronic databases, including PubMed, Scopus, IEEE Xplore, ArXiv, and was supplemented by reviewing materials from medical imaging conferences such as MICCAI, ISBI, and IPMI. Additionally, recognizing that medical image analysis is built directly on methods from general computer vision (e.g., CNNs for image classification), we also included a review of foundational and highly influential papers from that field in chronological order. To identify the earliest attempts to apply general-purpose CNNs to medical imaging, we used the Scopus engine, searching with medical-related keywords within papers that cited the original CNN publication. Furthermore, we examined the reference lists of key articles to identify additional sources. This initial process yielded over 1200 potential articles by 1 February 2025. Inclusion criteria required articles to be peer-reviewed,

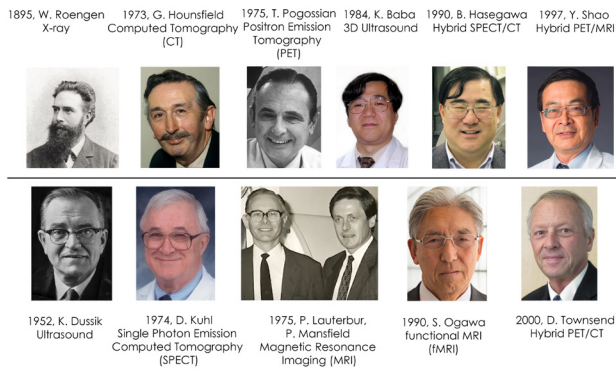


FIGURE 2. Key figures in medical imaging.

written in English, and available in full text. As detailed in the PRISMA flow diagram 1, we excluded duplicates, studies focusing solely on traditional machine learning, and preprints superseded by journal versions, resulting in the final selection of publications.

III. MEDICAL IMAGING MODALITIES

Since Wilhelm Roentgen's discovery of X-rays in 1895 [8], diagnostic imaging has evolved from simple radiography to an array of modalities including Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Ultrasound (US), and nuclear medicine (PET, SPECT). A chronological overview of these milestones is presented in Fig. 2. In modern practice, modality choice is dictated by the specific structural (anatomical) or functional (metabolic) information required. Table 1 summarizes the underlying physical principles, applications, and advantages, providing a comparative overview for applications.

A. X-RAY

Today, X-ray still remains one of the most widely utilized techniques in clinical practice because of its cost-effectiveness and rapid acquisition. Wilhelm Roentgen's discovery of X-rays remains one of the few scientific breakthroughs that can be precisely dated – the evening of November 8, 1895. His decision not to patent the discovery promoted widespread adoption, and within a year, the utility of X-rays was demonstrated in clinical settings, such as the detection of foreign bodies. Over the following decades, X-ray continued to evolve, driven by both technological progress and a better understanding of radiations effects. In the 1920s and 1930s, improvements in X-ray tubes, such as the development of the Coolidge tube in 1913, enabled precise control over the X-ray beam, which improved the efficiency of the imaging process [9]. Concurrently, progress in photographic plates (faster and more sensitive films) improved the quality of X-ray images while reducing exposure times. The 1930s also saw the development of fluoroscopy, a technique that provided real-time imaging by using continuous X-ray exposure and a fluorescent screen.

Concurrently, the development of fluoroscopy provided real-time visualization, which became essential for gastrointestinal and vascular studies [10]. A major shift occurred in the 1980s, with the development of charge-coupled device and complementary metal-oxide-semiconductor sensors, that allowed for the direct capture of digital X-ray images. This allowed digital radiographs to be easily stored, transmitted, and manipulated. The research in the 2010s prioritizes the development of low-dose, high-resolution detectors using new materials such as perovskites [11]. Unlike traditional scintillators (e.g., CsI:Tl), perovskites offer high sensitivity, while organic variants are explored for flexible, large-area fabrication [12]. To address persisting limitations in spatial resolution and optical crosstalk, the integration of high efficiency converting layers with metasurfaces represents a key direction for future detector architecture [13].

B. COMPUTED TOMOGRAPHY

As a successor to planar X-ray imaging, CT utilizes computational algorithms to reconstruct cross-sectional images of the body. CT enables the creation of slices through the body, which can then be stacked to form a 3D volume. This capability is essential for evaluating internal injuries and trauma, and has become a gold standard in cancer imaging and staging. The history of CT can be traced back to Allan Cormack, a physicist who derived solutions for reconstructing functions from line integrals in 1960s. Later, Godfrey Hounsfield developed practical instrumentation, leading to the first clinical CT scan of a patient on October 1, 1971 [14]. Early systems employed a translate-rotate geometry and required several minutes per scan, however, the shortly following introduction of windowing to handle dynamic range and computers for reconstruction proved the modality's clinical utility [15]. Cormack and Hounsfield shared the Nobel Prize in 1979 for these developments. Modality evolution progressed in the 1980s to improve temporal resolution and reduce artifacts. Second and third generation scanners introduced fan beams and rotate-rotate geometries, though early detector instability caused ring artifacts requiring calibration [16]. A major shift took place in 1987 with slip-ring design, which replaced cable connections to allow continuous rotation, resulting in reducing scan times to one second [17]. This enabled spiral CT in the early 1990s, allowing volumetric acquisition for imaging moving anatomical structures. The modern era of CT began in the late 1990s with the “slice race,” evolving from 4-slice systems to 320-slice Multidetector CT scanners that allow real-time whole-organ imaging [18]. The introduction of dual-source CT in 2006 revolutionized cardiac imaging by offering high temporal resolution [19]. Current clinical research focuses on spectral (dual-energy) CT for material decomposition and photon-counting CT, which improves spatial resolution to 150 μm while reducing electronic noise and radiation dose [20]. Simultaneously, research into Hierarchical Phase-Contrast Tomography is

breaking new ground by using synchrotron radiation to visualize anatomical structures down to the cellular level [21]

C. MAGNETIC RESONANCE IMAGING

Unlike X-ray-based modalities, Magnetic Resonance Imaging (MRI) offers superior soft tissue contrast without ionizing radiation, making it widely applicable for neurological, musculoskeletal, and cardiovascular diagnostics. Functional MRI (fMRI) maps brain activity via blood flow changes, while Dynamic Contrast-Enhanced MRI characterizes tumor microenvironments. Compared to CT, MRI offers superior soft tissue contrast, although with quite higher costs and longer imaging times. The foundation of Nuclear Magnetic Resonance was established in the 1940s by Bloch and Purcell. Medical application for imaging was demonstrated in 1973 by Paul Lauterbur, who employed magnetic field gradients to spatially encode signals, and Raymond Damadian, who showed that tissue relaxation times could differentiate tumors from normal tissue. Lauterbur and Mansfield were awarded the Nobel Prize in 2003. Following the first human imaging in by Lauterbur 1977, John Mallard and James Hutchison independently achieved similar results two years later [22]. By the mid-1980s, the modality was rebranded to MRI, and superconducting magnets replaced resistive ones, enabling higher field strengths and improved signal-to-noise ratios [23]. Innovations such as superconducting magnets, echo-planar imaging, and multi-slice acquisition dramatically reduced scan times, transforming MRI into a routine clinical tool by the 1980s [24]. By the 2000s, 3 T systems became the clinical standard, offering sub-millimeter resolution and enabling non-invasive angiography [25]. Current research focuses on Ultra-High Field systems operating at 7 T and 11.4 T to maximize resolution and signal-to-noise ratio [26], [27]. However, these higher field strengths introduce challenges regarding radiofrequency inhomogeneity and physiological effects such as vertigo [28]. Future generations aim to integrate synthetic CT generation from MRI data for radiation planning and refine 4D-MRI for real-time motion tracking of cardiac and respiratory cycles [29].

D. NUCLEAR IMAGING

Single-photon emission computed tomography (SPECT) and positron emission tomography (PET) visualize physiological processes and biological activities at the molecular level. Using radiopharmaceuticals, these modalities assess tissue biochemistry and organ function. Clinically, they are essential in oncology for tumor staging and monitoring and in neurology for diagnosing neurodegenerative disorders and mapping metabolic activity, and in cardiology for evaluating myocardial perfusion and viability. While sharing some conceptual similarities, SPECT and PET differ fundamentally in specific mechanisms of detecting radioactive emissions.

The development of SPECT traces back to Hal Anger's invention of the scintillation gamma camera in 1957. While

early systems faced limitations in efficiency and detection capabilities, the introduction of technetium-99m enabled the design of optimized detector crystals, leading to the scintillation camera superseding rectilinear scanners. Computational integration in the 1960s enabled the reconstruction of tomographic images from emission data. By the 1980s, the introduction of multi-headed gamma cameras significantly enhanced image quality and acquisition speed [30]

Developing in parallel with SPECT, Positron Emission Tomography (PET) traces its origins to the 1932 discovery of the positron. Practical imaging applications began in the 1950s with Brownell and Sweet, finalizing in the development of the first modern transaxial scanner by Tier Pogossian two decades later. A key development occurred in 1976 with Tatsuo Ido's development of 18F-fluorodeoxyglucose (FDG), which established PET as a primary tool for assessing glucose metabolism in oncology and neurology [31]. While early systems utilized filtered back-projection, iterative reconstruction methods became standard in the 1990s. This decade also marked a significant milestone with the introduction of hybrid PET/CT systems by Townsend and Nutt [32]. By the mid-2000s, standalone PET scanners were mainly replaced by hybrid systems (as discussed in Section III-F2), which integrate metabolic and anatomical data.

E. ULTRASOUND

Medical ultrasound, or sonography, is the most widely used imaging modality in modern medicine due to its affordability, portability, and lack of ionizing radiation. It serves a broad spectrum of clinical needs, ranging from routine obstetric monitoring and fetal development tracking to abdominal evaluations of the liver, kidneys, and pancreas. Additionally, ultrasound is essential for vascular assessments, such as detecting clots or guiding catheter placement, and diagnosing musculoskeletal conditions such as tendon tears.

The origins of clinical ultrasound lie in post-World War II research, influenced by military SONAR and Paul Langevin's piezoelectric studies. Early attempts in the late 1940s using "through transmission" were hindered by high attenuation, leading to the development of pulsed reflective ultrasound. This technology evolved from one-dimensional A-mode (amplitude) to two-dimensional B-mode (brightness) imaging, pioneered by Wild and Reid in the early 1950s. The subsequent introduction of M-mode (motion) in 1955 enabled the visualization of dynamic structures, such as cardiac valves. Commercial adoption accelerated in the 1960s with the development of articulating arm scanners, while the 1970s represented the transition to digital scan converters, which provided reproducible gray-scale imaging for the first time [33].

The transition to real-time in the mid-1970s marked a major shift, moving from mechanical scanning to multiple-element phased array systems that allowed electronic beam steering [34]. Although 3D ultrasound was reported as early as 1984 [35], it only became clinically viable

TABLE 1. Summary of key medical visualization methods. Resolution values are presented for latest systems available on the market.

	X-ray	CT	MRI	PET & SPECT	Ultrasound
Main applications	Skeleton fractures, luxation, bone disease	Oncology, trauma imaging, cardiovascular studies	Brain imaging, musculoskeletal disorders, soft tissue evaluation	Oncology, cardiology, neurology	Obstetrics, abdominal imaging, echocardiograph
Advantages	Widely available, high tissue penetration, low cost	High resolution, high tissue penetration	High resolution, no radiation, unlimited tissue penetration	High sensitivity, unlimited tissue penetration	No radiation, low cost
Limitations	Limited soft tissue contrast, ionizing radiation	Limited soft tissue contrast, ionizing radiation	Expensive, long acquisition time	Expensive, low resolution, radiation	Limited depth, low contrast, operator-dependent quality
Resolution	100 μm	125 μm	200 μm	0.6 mm	1 mm
Information type	Anatomical	Anatomical	Anatomical and functional	Functional and metabolic	Anatomical

in the 1990s with the introduction of matrix transducers and improved computational power [36]. In the 2000s, contrast-enhanced ultrasound using gas-filled microbubbles enabled tissue perfusion assessment. Inspired by optical super-resolution techniques [37], Ultrasound Localization Microscopy (ULM) emerged in the 2010s, achieving resolutions below 10 μm by tracking sparse microbubbles to visualize microvasculature [38]. Future developments are expected to focus on clinical applications of ULM [39], 4D imaging [40], and Robotic Ultrasound Systems to address operator variability and enhance reproducibility [41].

F. HYBRID IMAGING

The integration of multiple imaging modalities has been of significant interest in medical imaging since the 1960s, with the aim of providing a more comprehensive understanding of diseases [42]. In the early stages, this integration was primarily achieved through specialized software algorithms, primarily for brain imaging.

1) HYBRID SPECT/CT

Bruce Hasegawa pioneered hardware-based fusion, demonstrating the first simultaneous SPECT/CT prototype in 1990. This work laid the foundation for sequential hybrid scanners, commercialized as the GE Hawkeye in 1999. Although early systems had limitations in scan speed and resolution, they offered superior low-contrast detectability compared to conventional methods. The mid-2000s saw the introduction of multislice SPECT/CT (16 and 64-slice), followed by the adoption of iterative reconstruction in the 2010s, which reduced radiation doses by up to 80% [43]. A significant design shift occurred in 2017 with the introduction of 360-degree ring-shaped gantries utilizing CZT detectors. These systems enabled whole-body acquisition in under 20 minutes with improved contrast and sharpness [44]. Future directions will rely on new high-specificity tracers and sustainable production techniques to secure the $^{99\text{m}}\text{Tc}$ supply [45].

2) HYBRID PET/CT

Townsend reported the first integration of functional PET and anatomical CT in 2000, utilizing CT data for attenuation

correction. Commercial systems from General Electric and Siemens followed in 2001, effectively making PET only scanners obsolete by 2004. Following milestones included the introduction of full 3D reconstruction and fast lutetium oxyorthosilicate (LSO) crystals [46]. These enabled time-of-flight (TOF) capabilities, which significantly improved SNR and reduced acquisition times [47]. In 2015, continuous bed motion was introduced to improve sensitivity uniformity compared to traditional step modes [48]. A major hardware shift occurred in the 2010s with the replacement of photomultiplier tubes (PMTs) with solid-state silicon photomultipliers (SiPMs). SiPMs offer superior photon detection efficiency and timing resolution, directly translating to diagnostic improvement [49]. Current research targets sub-100 ps TOF timing through optimized crystal and edge-mounted SiPM configurations to further mitigate optical photon dispersion [50].

3) HYBRID PET/MR

PET/MR combines metabolic tracking with superior soft tissue contrast while eliminating CT associated ionizing radiation. The primary engineering challenge was integrating PET detectors into high magnetic field without mutual interference. Early attempts in 1997 using optical fibers to isolate photodetectors proved impractical due to signal loss. A clinical breakthrough occurred in 2010 with the Siemens Biograph mMR, which utilized avalanche photodiodes (APDs) tolerant to magnetic fields [51]. However, APDs lacked the timing resolution required for TOF. The subsequent adoption of SiPMs enabled the release of the first TOF-capable PET/MR system (United Imaging uPMR 790) in 2017, offering high spatial resolution and extended axial coverage [52]. Future research is focused on improving PET spatial resolution to below 1 mm [53].

IV. MACHINE LEARNING IN MEDICAL IMAGING

This section provides an overview of machine learning methods for tasks segmentation and classification tasks. Given the similarity between medical imaging and general computer vision methods, we begin by summarizing widely adopted computer vision models, including convolutional neural networks (CNNs) and Vision Transformers (ViTs),

followed by integration techniques presented while maintaining chronological order. We then examine architectures for segmentation, specifically U-Net variants and Transformer-based approaches. Finally, we summarize SOTA deep learning models, address current technical challenges, and discuss future research directions.

A. MACHINE LEARNING APPROACHES

The development of machine learning traces back to the nature-inspired neuron models of the 1940s and the perceptron in 1958, which laid the groundwork for the multilayer perceptron (MLP). In the medical domain, automated analysis appeared around 1960s and 1970s, initially relying on expert systems as CASNET [54] that used knowledge databases (if-else rules) for decision support. However, the fixed nature of these systems limited the ability to analyze complex image data.

By the mid-1980s, research shifted toward computer aided detection (CAD) systems designed specifically to assist radiologists. The first commercial CAD system, ImageChecker M1000 [55], was approved by the FDA in 1998. It utilized classical computer vision techniques with handcrafted features to detect microcalcifications. Yet, these systems suffered from high false positive rates and narrow rules, which eventually motivated the adoption of deep learning.

1) IMAGE CLASSIFICATION

In 1989, LeCun et al. [56] pioneered the application of neural networks trained via backpropagation for handwritten digit recognition. CNNs have become integral to image classification task for all domains, primarily due to their capacity to automatically learn hierarchical spatial features from image data. In the medical domain, CNNs began to integrate into CAD systems in the 1990s, for example, for the detection of lung nodules [57] and mammogram microcalcifications [58]. Although in 1990s CNNs showed promise, their full potential was initially constrained by data and computational limitations.

The landscape changed fundamentally in 2012 when Krizhevsky et al. [59] introduced AlexNet. By utilizing parallel GPU training and the large ImageNet-1k dataset (1.2 million images), AlexNet achieved a SOTA error rate of 37.5%, significantly outperforming the 45.7% error rate of the best handcrafted feature approach. This was the first demonstration of the feasibility of deep learning for complex classification tasks given sufficient data and compute. Medical researchers quickly adapted AlexNet for medical tasks: Roth et al. [60] for CT anatomy classification and Bar et al. [61] for X-ray pathology detection.

In 2014, Simonyan and Zisserman [62] proposed the Visual Geometry Group (VGG) network, utilizing repeating blocks of small (3×3) convolution filters to investigate the relationship between depth and performance. VGG-19 achieved SOTA results on ImageNet (25.5% error rate) and was

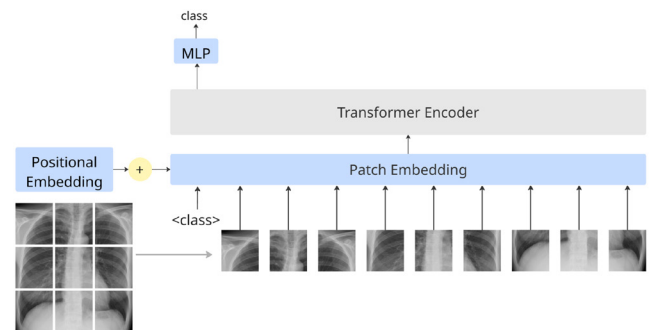


FIGURE 3. Visual Transformer (ViT) [82] overview.

subsequently applied by Akselrod-Ballin et al. [63] for tumor detection in mammography. In 2015, Szegedy et al. [64] introduced GoogLeNet (Inception), which addressed the challenge of kernel size selection by concatenating multiple convolutional branches. Achieving a 6.77% top-5 error rate on ImageNet, Inception was subsequently adapted for medical tasks, including breast cancer histopathology [65] and COVID-19 radiograph classification [66].

In 2016, He et al. [67] introduced the Residual Network (ResNet), utilizing residual connections to mitigate vanishing gradients in deep CNNs. ResNet-152 achieved SOTA results on ImageNet with a 19.38% top-1 error rate. The following year, Xie et al. [68] proposed ResNeXt, which introduced cardinality via grouped convolutions to reduce manual hyperparameter tuning, achieving a 19.1% error rate. Concurrently, Ishikawa et al. [69] applied residual learning to brain tumor classification.

In 2017, Huang et al. [70] introduced DenseNet, connecting layers in a feedforward manner using concatenation rather than summation. This design enhances feature reuse and parameter efficiency, achieving a 22.33% error rate. DenseNet was quickly adapted for pneumonia detection [71] and histology [72]. Subsequent architectures further improved different aspects of accuracy, efficiency and scalability. A chronological summary of foundational architectures and corresponding medical adaptations is provided in Table 2.

Parallel to this, in 2017, in the field of natural language processing (NLP), Vaswani et al. [92] proposed the Transformer architecture, which was specifically designed to address the key limitations of traditional sequence-to-sequence recurrent NLP models – the challenge of capturing long-range dependencies. Despite the obvious differences in the tasks between language and vision, a surprising convergence occurred in 2020 when Dosovitskiy et al. [82] introduced Vision Transformers (ViT), which changed the landscape of network architecture design. Apart from the initial patchify layer, as shown in Figure 3, which simply divides an image into a sequence of patches, ViT surprisingly introduces no image specific inductive bias and retains much of the original design of the NLP Transformers. Early medical adaptations appeared in 2021 for COVID-19 detection [84]. While

TABLE 2. Foundational deep learning architectures for image classification and their medical adaptations.

Year	Architecture (Citations)	Contribution Summary	Early Medical Adaptations
1989	LeNet-5 (18,563) [56]	Pioneered the first application of backpropagation in CNNs	Lung nodule detection (1995) [73] Microcalcifications detection (1996) [74]
2012	AlexNet (142,011) [59]	Pioneered deep CNNs for large-scale image classification	CT anatomy classification (2015) [60] Radiograph pathology detection (2015) [61]
2014	VGG (142,163) [62]	Demonstrated benefits of repeated 3×3 convolutional blocks	Tumor detection (2016) [63] Histology classification (2016) [75]
2015	Inception (66,063) [64]	Introduced multi-branch inception modules	Histopathology classification (2017) [65] Chest CT classification (2020) [66]
2016	ResNet (263,386) [67]	Proposed residual connections for deeper networks	Brain tumor classification (2016) [69] ECG classification (2017) [76]
2017	ResNeXt (14,494) [68]	Grouped convolutions for improved efficiency	Cancer histology (2018) [77] Slice selector (2018) [78]
2017	DenseNet (53,032) [70]	Dense inter-layer connections for feature reuse	Pneumonia detection (2017) [71] Histopathology magnification (2018) [72]
2019	EfficientNet (27,495) [79]	Compound scaling method for optimal model scaling	Skin lesion classification (2020) [80] Retinal fundus (2020) [81]
2020	Vision Transformer (78,054) [82]	Applied pure Transformer to sequence of image patches	COVID-19 detection (2021) [83] Ultrasound classification (2022) [84]
2022	ConvNeXt (7,648) [85]	Modernized ConvNets with Transformer	COVID-19 detection (2022) [86], [87]
2023	ConvNeXtV2 (1,057) [88]	Merged ConvNeXts with Masked Autoencoders	Skin cancer detection (2024) [89]
2024	InceptionNeXt (190) [90]	Decomposed large kernel convolutions for efficiency	Lung cancer detection (2025) [91]

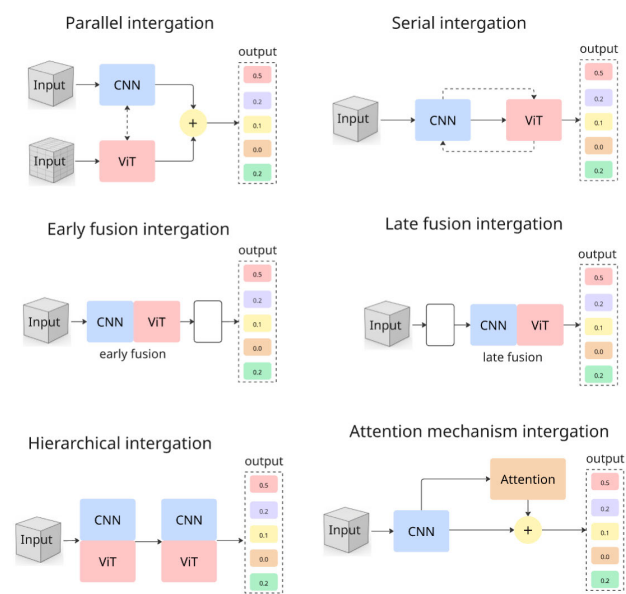


FIGURE 4. Approaches for CNN and vision transformer merging [93].

ViTs demonstrated superior scaling behavior and achieved SOTA ImageNet performance, the self-attention mechanism has quadratic complexity regarding input size, presenting memory and computation challenges for high resolution medical images.

To address this issue, hybrid architectures have been developed to combine the strengths of CNNs and ViTs. CNNs are excellent at extracting local (fine-grained) features through convolutions, but tend to struggle to capture long-range dependencies. Conversely, ViTs utilize self-attention mechanisms to model global context (long-range relationships), yet they are computationally heavy and less suited for local detail extraction. In general, merging CNNs and ViTs involves

parallel, serial, and hierarchical architectural integrations, as well as early-stage and late-stage feature fusions, and finally, the attention mechanism integration [93], as shown in Figure 4.

While the evolution from early CNNs to modern hybrid models has given significant performance gains, standard public benchmarks as ImageNet are approaching saturation. Future research is expected to move toward efficiency for resource-limited settings, reducing reliance on labeled data, and ensuring robustness in clinical applications.

2) IMAGE SEGMENTATION

In contrast to object classification, the semantic segmentation task aims at understanding image content on a pixel level. This involves labeling each pixel with a semantic category (e.g., predefined class). Standard CNNs for image classification use layers followed by a softmax function, which generates a one dimensional probability distribution, that is not compatible with two-dimensional pixel level classification. To address this, Long et al. [94] introduced the Fully Convolutional Network (FCN) in 2015. FCNs replace fully connected layers with convolutions to save spatial information, enabling dense prediction. This approach achieved SOTA performance on PASCAL VOC 2012 (62.2% mean IoU) and was adapted for CT anatomical structures in 2016 [95].

FCNs were intended for general-purpose tasks and require large volumes of data. Consequently, they could not be widely adopted in the medical field due to its primary constraint – limited datasets sizes. Thus, Ronneberger et al. developed U-Net [96], which quickly became the standard for biomedical segmentation after winning the 2015 ISBI Cell Tracking Challenge (0.920 IoU). The architecture utilizes an encoder and a decoder linked by skip connections. Such connections preserve high resolution details lost during

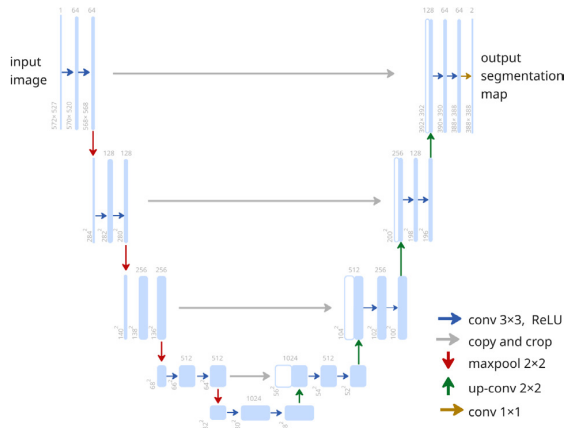


FIGURE 5. U-Net architecture [94] for medical semantic segmentation.

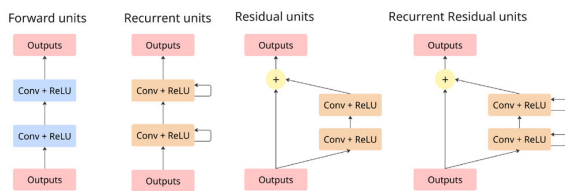


FIGURE 6. Different U-Nets convolutional units topologies [101].

downsampling, effectively combining low level spatial information with high-level semantic features. Despite its success, standard U-Net operates on 2D slices, presenting efficiency and consistency challenges when applied to volumetric data (e.g., CT, MRI) where adjacent slices share significant information.

In 2016, Cicek et al. [97] addressed this by extending U-Net operations to 3D. The model achieved an average IoU of 0.863 on Xenopus kidney data set and demonstrated reliability across brain and abdominal segmentation tasks [98]. Concurrently, Milletari et al. [99] introduced V-Net for prostate MRI. Distinct from U-Net, V-Net adds residual blocks (Figure 6) and introduced the Dice loss function to handle foreground-background voxel imbalance. Using nonlinear augmentations, it achieved a 0.869 Dice score on the PROMISE2012 dataset [100].

In 2018, Alom et al. [101] introduced an approach to combine the architectural strengths of RCNNs and ResNets into a family of U-Net based segmentation networks. Specifically, for the first network, termed RU-Net, the authors substituted the conventional forward convolutional units of the U-Net architecture with recurrent convolutional layers (RCLs) adapted from RCNNs. This change was designed to promote the aggregation of features, leading to improved segmentation results. In the second network, abbreviated as R2U-Net, the RCLs were further modified by the residual units inspired by ResNet. Both models outperformed standard and residual U-Nets across retinal datasets (e.g., DRIVE 0.8171 F1-score). Concurrently, Oktay et al. [102] were the first to propose including an attention gate in the

standard 3D U-Net in, enabling the network to focus on target structures of varying shapes and sizes. Evaluation of Attention-UNet on CT abdominal segmentation (CT-150) yielded new SOTA Dice score of 0.840. Interestingly, their attempts to add residual connections did not provided measurable performance gains.

Subsequent research addressed uncertainty and semi-supervised learning. In 2019, Ye et al. [103] proposed an uncertainty-aware, self-ensembling mean teacher framework for 3D MRI left atrium segmentation, achieving 88.88% DSC [104] by efficiently using unlabeled data. In 2020, Mehrtaash et al. [105] assessed uncertainty estimation for medical image segmentation using 2D U-Nets on MRI images, as these models tend to produce overconfident predictions. The results showed that for baseline (single) models, the cross-entropy loss outperformed the Dice loss in terms of uncertainty estimation, although it fell short in segmentation quality. More importantly, the study strongly demonstrated that ensembling significantly calibrates the model's confidence.

One of the first successful attempts to merge CNNs with transformers for segmentation (an early fusion type of integration) was reported by Chen et al. [106] in 2021 in the form of TransUNet. This hybrid architecture utilizes a Transformer to encode CNN feature patches for global context, which are then combined with high resolution CNN maps in the decoder for localization. The model achieved 0.775 DSC on the Synapse multi-organ CT dataset [107], surpassing V-Net and attention-gated U-Net.

In the following year 2022 Cao et al. [108] proposed Swin-UNet, the first Transformer-only SOTA architecture based on UNet for medical image segmentation. Eliminating convolutional layers, it employs hierarchical Swin Transformer blocks [109] within a symmetric encoder-decoder structure. Utilizing shifted windows and patch-expanding layers to capture global context, Swin-UNet outperformed hybrid approaches (e.g., TransUNet), achieving 0.791 DSC on the Synapse dataset [107] and 0.90 DSC on ACDC [110].

In parallel, Zhang et al. [111] proposed TransFuse, which separately processes Transformer and CNN branches before selectively fusing multi-scale features to capture both local and global context. This parallel integration achieved 0.920 DSC on the Kvasir polyp dataset [112]. For 3D volumetric tasks, Hatamizadeh et al. [113] introduced UNETR, utilizing a Swin Transformer encoder connected to a CNN decoder via multi-resolution skip connections. This hierarchical fusion demonstrated strong performance on BraTS 2021 [114] (0.926 DSC for Whole Tumor). Addressing computational efficiency in 2023, Xu et al. [115] proposed DCSAU-Net, employing compact split-attention blocks to enhance feature representation across channels. This method achieved SOTA results across multiple benchmarks, including CVC-ClinicDB [116] (0.916 F1) and ISIC-2018 [117].

In 2024, Lan et al. [120] proposed BRAU-Net++, which incorporates sparse dynamic attention via BiFormer

TABLE 3. Evolution of key image segmentation architectures.

Year	Authors	Architecture (Citations)	Performance	Contribution Summary
2015	Long et al. [94]	FCN (47,831)	PASCAL VOC 2012: 0.622 IoU	Adapted convolutional networks for segmentation
2015	Ronneberger et al. [96]	U-Net (108,735)	ISBI 2015: 0.920 IoU	Pioneered U-shaped architecture with skip connections
2016	Cicek et al. [97]	3D U-Net (8,825)	Xenopus Kidney: 0.863 IoU	Extended U-Net to 3D for volumetric data segmentation
2016	Milletari et al. [99]	V-Net (12,103)	PROMISE2012: 0.869 DSC	Introduced residual blocks and Dice loss
2018	Alom et al. [101]	R2U-Net (1,591)	DRIVE: 0.817 F1	Combined recurrent and residual units in U-Net
2018	Oktay et al. [102]	Attention-UNet (7,749)	CT-150: 0.840 DSC	Integrated mechanism as attention gates in U-Net
2021	Isensee et al. [118]	nnU-Net (5,457)	MSD: 0.790 DSC	Proposed automatic network configuration
2021	Chen et al. [119]	TransUNet (5,811)	Synapse CT: 0.775	CNN-Transformer fusion with global context
2022	Cao et al. [108]	Swin-Unet (4,283)	Synapse CT: 0.791 DSC	First pure Transformer-based U-Net based architecture
2022	Zhang et al. [111]	TransFuse (1,290)	Polyp: 0.920 DSC	Parallel CNN-Transformer fusion with multi-scale features
2022	Hatamizadeh et al. [113]	UNETR (1,465)	BraTS 2021 (WT): 0.926 DSC	Transformer encoder with CNN decoder for 3D data
2023	Xu et al. [115]	DCSAU-Net (246)	CVC-ClinicDB: 0.920 F1	Compact split-attention blocks for multi-scale features
2024	Lan et al. [120]	BRAU-Net++ (26)	Synapse: 0.825 DSC	Dynamic sparse attention with bi-level routing
2025	Peng et al. [121]	U-NET V2 (193)	CVC-ClinicDB: 0.944 DSC	Redesigned skip connections

blocks [122] to efficiently model long-range dependencies. The architecture utilizes depth-wise convolutions for implicit positional encoding within a U-shaped structure. Evaluations on Synapse [107], ISIC-2018 [117], and CVC-ClinicDB [116] demonstrated SOTA performance with DSC scores of 0.825, 0.923, and 0.929, respectively.

More recently, another paradigm has emerged in the form of diffusion probabilistic models, which reformulate segmentation as a generative task. Initially, large diffusion models such as Stable Diffusion and Imagen showed unprecedented image generation capability. Applied for segmentation, such models instead of directly predicting a segmentation map, these models learn to reverse a diffusion process, iteratively denoising a random noise map into a final segmentation mask. Amit et al. [123] introduced SegDiff, which conditions a U-Net on input features and averages generated masks to improve calibration. Adapting this to the medical domain, Wu et al. [124] proposed MedSegDiff, employing dynamic conditional encoding and a Fourier-domain frequency parser to suppress noise in low-contrast regions. Its successor, MedSegDiff-V2 [125], successfully integrated Transformers via anchor conditioning and spectrum-space fusion, achieving SOTA results across 20 tasks. For volumetric data, Xing et al. [126] developed Diff-UNet, which embeds diffusion within a U-Net and utilizes a step-uncertainty fusion strategy to weight predictions throughout the denoising process.

In contrast to this line of work, recent progress in natural image segmentation have led to the development of foundational segmentation models. In 2023 Kirillov et al. [127] introduced the Segment Anything Model (SAM), trained on 1 billion masks for zero-shot promptable segmentation, followed by the video-optimized SAM 2 [128]. In the medical domain, Cheng et al. [129] developed SAM-Med2D by fine-tuning SAM on 4.6 million images, demonstrating superior generalization. Similarly, Ma et al. [130] proposed MedSAM, utilizing 1.6 million masks across various modalities to outperform specialist U-Nets. More recently, Archit et al. [131] introduced MedicoSAM, concluding that

while foundational models achieve SOTA performance in interactive bounding-box scenarios, specialist models often remain superior for fully automated semantic segmentation due to SAM's class-agnostic learning objective.

Lastly, as the image segmentation field has rapidly evolved since 2015, beginning with the U-Net [96] architecture. Key improvements have involved extending the U-Net to 3D (3D U-Net [97], V-Net [99]) and integrated components such as residual connections, attention mechanisms, and recurrent units. More recently, the rise of Transformer architectures, for various hybrid CNN-Transformer models (TransUNet [119], UNETR [113], TransFuse [111], BRAU-Net++ [120]) and pure Transformer designs (Swin-Unet [108]), has significantly improved model performance. Concurrently, large-scale foundational models such as SAM [127] and its medical adaptations (MedSAM [130], MedicoSAM [131]) are introducing a new paradigm focused on zero-shot promptable segmentation, yet specialized domain expert models still excel in specific tasks. We expect future research to focus on refining hybrid architectures and the eventual arrival of foundational models capable of competing with domain expert models. Another line of work lies in semi-supervised learning to address data gathering and annotation challenges, as well as ensuring model robustness and generalizability for reliable deployment in real-world applications.

B. MEDICAL IMAGING APPLICATIONS

This section provides a selective overview of machine learning methods applied to clinically significant tasks in medical image analysis, including segmentation and classification across different anatomical structures as well as imaging modalities. We examine segmentation tasks ranging from whole-body tumor and organ delineation to specialized applications in head and neck, cardiac, ophthalmic, and microscopic imaging. Furthermore, we review classification approaches for pathologies including diabetic retinopathy, Parkinson's disease, COVID-19, and various cancers (prostate, gastric, skin, and breast). For each application, we summarize SOTA performance, addressing real world

TABLE 4. Overview of datasets and top-performing methods for medical segmentation.

Dataset	Description & Data licence	Method	Performance
AutoPET (2022) [132]	Segmentation of tumor lesions on whole body PET/CT, 1014 cases, 1 category body tumor segmentation. CC BY NC	DynUNet [133]	56.52% DSC
		Semi-supervised Unet [134]	92.28% DSC
		UNet [135]	92.81% DSC
TotalSegmentator [136]	Segmentation of anatomical structures, CT, 1228 Cases, 117 categories of whole body. Apache-2.0 licence	STU-Net [137]	90.06% DSC
		nnU-Net [136]	94.30% DSC
TotalSegmentator MRI [138]	Segmentation of major anatomical structures, MRI, 298 cases, 56 categories of whole body organ segmentation. Apache 2.0 licence	nnU-Net [138]	83.90% DSC
Medical Segmentation Decathlon (MSD) [139]	Segmentation of major anatomical structures, CT and MRI, 2633 cases, 10 categories. CC BY SA licence	nnU-Net [139]	79.00% Median DSC
Computed Tomography Organs (CT-ORG) [140]	Segmentation of organs: lungs, bones, liver, kidneys, and bladder, 131 CT cases and 9 PET-CT cases, 5 categories. CC BY licence	FSS-ULivR [141]	91.72% Mean DSC
Brain Tumor Segmentation (BraTS21) [142]	Segmentation of brain tumor, MRI, 1241 cases, 3 categories of glioma: Enhancing Tumor (ET), Whole Tumor (WT), Tumor Core (TC). CC BY NC licence	Swin UNETR [113]	Mean 88.97% DSC
		MedNeXt [143]	91.48% DSC
		GAN + nnU-Net [144]	Mean 91.74% DSC
ToothFairy [145]	Segmentation of maxillofacial structures, CBCT, 443 cases, 1 category. CC BY SA licence	MATAP [146]	80.60% DSC
ToothFairy 2 [147]	Segmentation of maxillofacial structures, teeth and maxillary bone, CBCT, 493 cases, 42 categories. CC BY-SA licence	UMamba [147]	85.05% DSC
Thyroid Nodule (TN3K) [148]	Segmentation of thyroid nodule, 2D Ultrasound, 3585 Cases, 1 category. MIT licence	TRFE+ [149]	83.30% DSC
		DPAM-UNet++ [150]	74.51% IoU
Multi-Modality Whole Heart Segmentation (MM-WHS) [151]	Segmentation of heart structures (myocardium, left atrium, left ventricle, right atrium, right ventricle, ascending aorta, pulmonary artery), 3D CT and MRI, 120 cases, 7 categories. CC BY licence	SIFA [152]	73.00% DSC
		Semi-supervised framework [153]	79.83% DSC
Cardiac ultrasound video dataset (CardiacUDA) [154]	Segmentation of heart structures: pulmonary artery (PA), left ventricle (LV), left atrium (LA), right atrium (RA), and right ventricle (RV), video 2D Ultrasound, 992 cases, 5 categories. Apache 2.0 licence	GraphEcho [154]	76.60% DSC
		M ³ -UDA [155]	94% mAP
Myocardial pathology segmentation (MyoPS) [156], [157]	Segmentation of heart structures: scar, edema, normal myocardium, and left and right ventricular blood pools, MRI, 45 cases, 5 categories. CCBY licence	AWSnet [158]	Mean 70.65% DSC
		MyoPS-Net [157]	Mean 77.90% DSC
Whole Heart Segmentation segmentation of MR (HVS MR-2.0) [159]	Segmentation of heart structures (left ventricle (LV), right ventricle (RV), left atrium (LA), right atrium (RA), aorta (AO), pulmonary artery (PA), superior vena cava (SVC) and inferior vena cava (IVC)), 3D MR, 60 cases, includes masks of optional extents of the great vessels. CCBY licence.	Hybrid CNN [160]	Mean 86.46% DSC
Cataract-1K [161]	Video surgery segmentation, 1000 Cases, 13 categories of surgical phases, 2 categories of anomaly detection, 12 categories of semantic segmentation. CC BY licence.	VGG + DeepPyramid [161]	Mean 89.17% DSC

challenges such as class imbalance and imaging variability. The section also outlines data licences for open access datasets and discusses emerging trends. Table 4 provides a detailed performance summary, listing the specific datasets, task descriptions, network architectures, and reported performance metrics for each application.

1) MEDICAL SEGMENTATION

Whole-body segmentation in medical imaging is crucial for oncology and surgical planning, yet it faces challenges in terms of accuracy and efficiency. To address the labor intensive nature of tumor lesion segmentation, Gatidis et al. [132] proposed the autoPET challenge for whole-body FDG PET/CT. For this task, Hadlich et al. [133] employed a DynUNet modification, focusing on data augmentation and hyperparameter tuning. Peng et al. [134] introduced a false positive reduction network utilizing a self-supervised pre-trained ResNet-50 encoder to improve candidate tumor regions. The current SOTA solution by Ye et al. [135] utilized a vanilla U-Net approach optimized via ablation analysis on

input shape, achieving robust performance without complex post-processing.

Similarly to tumor segmentation in whole-body PET/CT scans, segmentation of anatomical structures in more commonly used whole-body CT scans is essential. For instance, in routine clinical practice, CT segmentation has already found use in areas such as surgical planning and radiotherapy. However, manual annotation remains a time-consuming task that requires a high level of expertise. To address these challenges, Wasserthal et al. [136] introduced the TotalSegmentator dataset covering 117 anatomical categories. They applied the nnU-Net framework to establish a SOTA baseline. Addressing scalability, Huang et al. [137] proposed the Scalable and Transferable U-Net (STU-Net), ranging up to 1.4 billion parameters. Their work demonstrated that simultaneously scaling both depth and width is an effective strategy for large-scale pretraining, though challenges persist in differentiating neighborhood structures. Extending TotalSegmentator to MRI, Dantonoli et al. [138] introduced TotalSegmentator MRI to handle varying resolutions and contrasts. The authors applied the nnU-Net model, which

demonstrated robust performance, although difficulties remain with lower resolution or highly anisotropic sequences.

The Medical Segmentation Decathlon (MSD) challenge, introduced by Antonelli et al. [139], was gathered to address one of the main challenges in medical image analysis – the development of a general purpose segmentation model capable of performing robustly across a variety of clinical tasks, modalities, and anatomical structures. Designed to present real-world difficulties including class imbalance, small target objects, and high data variability, the challenge is used as a benchmark for model adaptability. A key outcome was the validation of the nnU-Net framework [118], which emerged as the winning solution.

The challenge of segmenting multiple organs simultaneously from CT scans, essential for comprehensive diagnostic and surgical planning, is a key focus of the Computed Tomography Organs (CT-ORG) dataset [140]. To mitigate the reliance on large datasets, Debnath et al. [141] proposed FSS-ULivR, a few-shot segmentation framework. The architecture integrates a ResNet-based encoder with squeeze-and-excitation modules, a transformer for feature refinement, and an attention gated decoder. FSS-ULivR achieved strong performance, surpassing previous SOTA methods such as TransUNet and Swin-Unet, demonstrating generalizability in limited-data scenarios.

Magnetic resonance imaging of the brain is the gold standard in the clinical diagnosis and management of brain tumors, particularly gliomas, which are known for their aggressive behavior and complex morphology. To address the variability, the Brain Tumor Segmentation Challenge (BraTS) provides a benchmark for evaluating SOTA approaches. The BraTS 2021 dataset [142] standardized multi-modal segmentation, while subsequent iterations addressed missing data synthesis [162] and post-treatment resection cavity segmentation [163]. Hatamizadeh et al. [113] proposed Swin UNETR, reformulating 3D segmentation as a sequence to sequence task using a hierarchical Swin transformer encoder connected to a CNN decoder, ranking among top-performing approaches. Roy et al. [143] introduced MedNeXt, a fully ConvNeXt 3D network utilizing iterative kernel upscaling and compound scaling to preserve semantic richness. Furthermore, Ferreira et al. [144] utilized generative adversarial networks (GAN) to augment training data for an nnU-Net baseline, achieving SOTA results across all tumor regions.

Segmentation of the inferior alveolar canal (IAC) in cone beam computed tomography (CBCT) scans is crucial for various dental and maxillofacial surgical applications, as it helps to prevent irreversible nerve damage during procedures. The Tooth Fairy Challenge [145] addresses this using sparse and dense annotations. Lumetti et al. [146] proposed MATAP, a 3D U-Net variant incorporating a memory-augmented Transformer in the bottleneck and absolute positional tokens, which demonstrated top performance. The subsequent Tooth Fairy 2 [147] expanded the task to include mandible, teeth, and pharynx structures. Benchmarking against architectures

as nnU-Net and TransU-Net, the authors concluded that UMamba [164] achieved superior results.

The segmentation of thyroid nodules on ultrasound is vital for the early detection and diagnosis of thyroid cancer, offering significant clinical value. Automating this process can ease the burden on clinicians who are currently relying on manual annotations and high-expertise. For this Gong et al. [148] introduced the TN3K dataset to benchmark algorithms against real-life variability in nodule views and intensities. Gong et al. [149] subsequently proposed TRFE+, a multitask framework utilizing adaptive region priors to simultaneously segment the thyroid gland and nodule. By constraining localization to the thyroid region, the method effectively reduced false positives. Dong et al. [150] introduced DPAM-UNet++, which integrates a dual path attention mechanism into UNet++ skip connections. Trained with a composite loss function for improved edge segmentation, the model demonstrated SOTA performance.

Cardiac segmentation is critical for diagnosing cardiovascular diseases by enabling precise quantification of cardiac structures (e.g., ventricles, atria, myocardium) for assessing heart function. One of the key benchmarks for this task is the Multi-Modality Whole Heart Segmentation (MMWHS) dataset [151]. Chen et al. [152] proposed Synergistic Image and Feature Adaptation (SIFA) to address domain shift, specifically between MRI and CT scans. SIFA employs GANs with cycle consistency and feature adaptation via discriminators to enhance domain invariance. By optimizing a shared feature encoder end-to-end, the method significantly improves segmentation performance without requiring target domain annotations. Addressing data scarcity, Basak et al. [153] introduced a semi-supervised learning method utilizing a ResNet50 encoder and consistency regularization through interpolation-based mixing. This approach enforces consistency between interpolated images and segmentation maps to leverage unlabeled data effectively. Evaluated on MMWHS with limited labeled volumes, the method outperforms other SOTA semi-supervised techniques. Regarding congenital heart disease (CHD), where anatomical connections are often atypical, Pace et al. [159] assembled the HVSMR-2.0 dataset from cardiovascular magnetic resonance (CMR) images. The dataset provides masks for eight heart structures and the great vessels, enabling robust algorithm comparison for surgical planning in complex CHD cases.

The CARDIACUDA dataset [154] addresses the challenge of domain gaps in cardiac structure segmentation from echocardiogram videos across different clinical centers. To mitigate these shifts, Yang et al. [154] proposed GraphEcho, a graph unsupervised domain adaptation (UDA) method. This approach utilizes Spatial-wise cross domain Graph Matching to align class-specific representations and Temporal Cycle Consistency to model heartbeat cyclicity, essentially using prior knowledge to improve intra-class similarity. Alternatively, Pu et al. [155] proposed M^3 -UDA, a multi-matching framework. This method incorporates

histogram matching for pixel transformation, sub-structure matching to bridge angular gaps, and global-structure matching to align topological knowledge between source and target domains, demonstrating strong performance. Regarding myocardial viability, Li et al. [165] introduced the MyoPS dataset, consisting paired multi-sequence CMR images (bSSFP, LGE, T2-weighted) to leverage complementary information for pathology segmentation. Wang et al. [158] proposed AWSnet, which utilizes a coarse-to-fine strategy where a coarse model provides shape priors to guide fine segmentation. The network employs reinforcement learning for auto-weighted supervision and pixel-wise attention to extract pathological features. Similarly, Qiu et al. [157] introduced MyoPS-Net, an end-to-end network designed to fuse cross-modal features. It incorporates a custom myocardium consistency module to regularize segmentation output and utilizes dedicated branches for targeting specific pathologies.

2) MEDICAL CLASSIFICATION

The APTOS 2019 dataset was collected for the detection and grading of diabetic retinopathy (DR) that considered one of the most significant vision concerns worldwide, mainly due to its impact on causing vision loss among many people with diabetes. The dataset supports both binary and multi-class grading tasks, presenting a challenge due to class imbalance. Regarding top-performing approaches, Karkera et al. [166] proposed an ensemble of transformer-based networks (ViT, DeiT, CaiT, and BEiT) utilizing weighted mean and majority voting schemes. The weighted mean approach yielded superior results, achieving near-perfect agreement with human grading and demonstrating robustness against dataset imbalance. Alternatively, Shakibania et al. [167] introduced a dual-branch network combining ResNet50 and EfficientNetB0, fine-tuned via transfer learning. This model demonstrated strong alignment with expert assessments in both binary detection and multi-class grading tasks.

The Cataract-1K proposed by Ghamsarian et. al [161] shifts focus on another relevant challenge in eye care: cataract surgery. As one of the most frequently performed eye procedures worldwide, cataract surgery is critical to restore vision in millions affected by lens opacification, one of the leading causes of blindness [195]. The dataset [161] addresses cataract surgery, enabling automated video analysis. The dataset includes frame level surgical phase labels, masks for anatomical structures and instruments, and subsets targeting irregularities such as pupil contraction. To validate the annotations, the authors evaluated several models for semantic segmentation. Among these, the DeepPyramid network utilizing a VGG16 backbone demonstrated superior performance across anatomical structures and instruments compared to other approaches.

The classification of otoscopic images is crucial for diagnosing ear conditions that range from common infections to more serious pathologies. To support otoscopic image

classification, Camalan et al. [169] introduced the OtoMatch dataset. Addressing label scarcity, Rahman et al. [170] proposed the Weakly-supervised Chained Distillation Network (WeCKD). This framework employs a chained process where a sequence of models is trained on different data subsets, with each learning from previous predictions. The method achieved competitive results using only a fraction of the labeled data required by fully supervised approaches.

The evaluation of bone marrow (BM) cytomorphology is essential for diagnosing hematological diseases, such as leukemias and myelodysplastic syndromes. However, manual analysis remains labor intensive and prone to observer variability. For this task Matek et al. [171] curated the BM cytomorphology dataset, currently the largest collection of its kind, reflecting real-world class imbalance. The authors established a baseline using a ResNeXt-50. While the model outperformed feature-based approaches and achieved high precision for major classes, it struggled with very rare categories, highlighting the common challenge of long-tail distributions in medical datasets.

Classification of cervical cytology images is central for improving early detection of cervical cancer, that is the fourth most common cancer in women worldwide after breast, colorectal and lung cancers. The Herlev Pap Smear dataset [172] serves as a notable benchmark for this task. Sornapudi et al. [173] utilized a graph-based detection pipeline with a fine-tuned VGG-19, demonstrating robustness in handling cell overlaps. Alternatively, Basak et al. [174] achieved SOTA performance by fusing features from multiple CNNs using PCA and the Gray Wolf Optimizer. For larger-scale analysis, Plissiti et al. [175] introduced the SIPAKMED dataset, containing both isolated and clustered cells. Mana et al. [176] proposed an ensemble of pre-trained CNNs integrated via fuzzy rank fusion, yielding superior accuracy in both binary and multi-class settings compared to individual models.

The diagnosis of prostate cancer is fully based on the Gleason scoring system, which evaluates histological patterns to determine the prognosis of the disease. The SICAPv2 dataset [177] addresses this by providing pixel-level Gleason annotations on whole slide images. Silva-Rodriguez et al. proposed a simple CNN trained from scratch that outperformed fine-tuned architectures in patch-level grading. Their framework also successfully addressed the detection of cribriform patterns and demonstrated superior alignment with expert decision-making at the biopsy level. Bian et al. [178] introduced a mixed-supervision transformer utilizing multiple instance learning to handle noisy annotations. This approach achieved SOTA performance for slide-level classification, with visual analysis confirming precise localization of Gleason patterns. For larger scale analysis, the PANDA challenge dataset [179] serves as the most extensive multi-site collection of prostate biopsies. Algorithms trained on this benchmark have demonstrated pathologist-level concordance and high sensitivity for tumor detection across diverse validation sets.

TABLE 5. Overview of datasets and methods for medical classification.

Dataset	Description & Data licence	Method	Performance
APTOS 2019 [168]	Classification of eye fundus, 3662 cases, 5 categories: no DR, mild, moderate, severe, proliferative DR. Custom licence.	ResNet50 + EfficientNetB0 [167]	89.60% accuracy
OtoMatch [169]	Classification of middle ear infection, 454 cases, 3 categories. CCBY 4.0 licence.	ViT Ensemble [166]	94.63% accuracy
Bone marrow (BR) cytology [171]	Classification of bone marrow cell cytology, 171,375 cases, 21 categories. CCBY 4.0 licence.	WeCKD [170]	88.99% F1 score
Herlev Pap Smear (HERlev) [172]	Classification of cervical cytology, Microscopic Imaging, 917 cases, 7 cell categories. CC BY 4.0 licence.	ResNeXt-50 [171]	61.00% precision
SIPAKMED [175]	Classification of cervical cytology, Microscopic Imaging, 4049 cases, 5 cell categories: superficial-intermediate, parabasal, koilocytotic, dyskeratotic, metaplastic. CC BY 4.0 licence.	VGG-19 [173]	88.86% accuracy
SICAPv2 [177]	Classification of prostate cancer cells, 4 categories, 182 cases. CC BY 4.0 licence	PCA + CNNs Ensemble [174]	98.32% accuracy
PANDA [179]	Classification of prostate Gleason grading, 5 categories, 10,616 cases. CC BY NC licence	Fuzzy logic CNNs Ensemble [176]	98.55% accuracy (2-cls), 95.43% accuracy (5-cls)
Parkinson's Progression Markers Initiative (PPMI) [181]	Classification of Parkinson's disease, 2 categories, 645 cases, SPECT modality. Custom licence.	PCA + CNNs Ensemble [174]	97.87% accuracy (5-cls)
COVID-CT [183]	Classification of coronavirus infection, 2 categories, 746 cases, CT modality. CC BY 4.0 licence.	ConvNet [177]	82.00% AUC
COVIDGR [184]	Classification of coronavirus infection, 4 categories, 852 cases, X-ray modality. CC BY 4.0 licence.	Mixed-supervision Transformer [178]	94.29% AUC
SIIM-FISABIO-RSN [185]	Classification of coronavirus infection, 4 categories: negative, typical appearance, indeterminate appearance and atypical appearance, 7597 cases, X-ray modality. CC-BY NC licence.	UNI [180]	75.70% accuracy
GasHisSDB [187]	Classification of Gastric cancer, 2 categories, 245,196 cases, microscopic imaging. CC BY 4.0 licence.	Fuzzy logic CNNs Ensemble [182]	98.45% accuracy
SkinCancer [189]	Classification of skin cancer, 16 categories, 129,364 cases, microscopic imaging. Custom licence.	Self-supervised DenseNet-169 [183]	89.10% accuracy
SLICE-3 (ISIC 2024) [190]	Classification of skin cancer, 2 categories, 401,059 cases, total body photographs. CC-BY NC.	COVID-SDNet [184]	97.72% accuracy
Camelyon16 [192]	Classification of metastatic breast cancer, 2 categories, 327,680 cases, microscopic imaging. CC-BY 1.0 licence.	VGG-19 [186]	91.00% accuracy
		VGG16 [187]	96.47% accuracy
		CNNs Ensemble [188]	99.20% accuracy
		EfficientNetV2 [189]	98.70% accuracy
		MobileNetV2 [191]	97.70% accuracy
		P4M-DenseNe [193]	84.00 % FROC
		Inception V3 [194]	88.50 % FROC

Parkinson's disease (PD) is the second most prevalent neurodegenerative disorder after Alzheimer's disease, currently it does not have a definitive diagnostic test. Noninvasive imaging techniques, mainly (PET) scans, can assist in the diagnostic process. The Parkinson's Progression Markers Initiative (PPMI) [181] utilizes PET imaging to identify biomarkers of PD progression. To address classification, Kurmi et al. [182] proposed an ensemble approach integrating four CNN models (VGG16, ResNet50, Inception-V3, and Xception). A fuzzy fusion logic-based ensemble was applied to generate fuzzy ranks for each class, yielding robust performance metrics.

Coronavirus Disease 2019 (COVID-19) was a highly contagious infectious disease, and one of the significant challenges in controlling its spread was the insufficient availability of diagnostic tests. During the peak of the COVID-19 outbreak, test kits were in severe shortage, forcing hospitals to use alternative diagnostic methods, such as X-ray and CT scans. In response to this problem, Yang et al. [183] curated the COVID-CT dataset to address diagnostic shortages. They developed a method employing multi-task and self-supervised learning techniques to integrate lung and lesion masks, achieving clinically useful accuracy. Similarly, Tabik et al. [184] curated the COVIDGR-1.0 dataset, categorized into four severity levels. They proposed the COVID Smart Data-based Network (COVID-SDNet), which utilizes data augmentation and a custom CNN to achieve consistent performance across severity levels. One of the largest

datasets was curated by Lakhani et al. [185], consisting over 10,000 chest radiographs including controls with various pulmonary pathologies. To address noisy labels and high sample variance, Zhang et al. [196] proposed a collaborative unsupervised domain adaptation framework utilizing peer networks and a noise co-adaptation layer, demonstrating superior results over existing unsupervised methods.

Gastric cancer is the fifth most commonly diagnosed cancer [197]. The gold standard for diagnosing gastric cancer involves the microscopic examination of paraffin-embedded tissue sections obtained from suspected cancerous regions. Sung et al. [198] introduced the Gastric Histopathology Sub-size Image Database (GasHisSDB). The authors evaluated several methods, noting that while VGG16 achieved high accuracy, it was less efficient than ResNet50 or ViT. Yong et al. [188] proposed an unweighted averaging ensemble comprising MobileNet, DenseNet, EfficientNet, Inception V3, and Xception. Evaluation revealed that the ensemble approach achieved superior detection accuracy compared to individual base models.

Skin cancer diagnosis requires differentiating between various non-melanotic and melanotic neoplasms [199]. Kriegsmann et al. [189] curated the SkinCancer dataset to aid in differentiating structures such as basal cell carcinoma, squamous cell carcinoma, and melanoma. Employing an EfficientNetV2 model, they demonstrated the feasibility of automated differentiation with high accuracy. To address limitations such as selection bias and lack of standardization,

Kurtansky et al. [190] curated SLICE-3D using 3D Total Body Photographs. Utilizing this dataset, Mrabet et al. [191] evaluated various SOTA CNNs, reporting that MobileNetV2 demonstrated the strongest performance.

While these supervised methods achieve excellent accuracy, they depend on large manually annotated datasets, which are expensive and not always accessible. To mitigate this reliance on labeled data, Rahman et al. [200] introduced an unsupervised framework combining GAN-based image enhancement, feature extraction, dimensionality reduction, and k-means clustering. Evaluated on the ISIC 2019 dataset, the approach surpassed other clustering algorithms and methods utilizing features from SOTA supervised models.

Accurate breast cancer staging is an essential task performed by pathologists to guide clinical management, a process critically dependent on assessing the extent of cancer spread through histopathological analysis of sentinel axillary lymph nodes (SLNs). Nevertheless, the sensitivity of SLN assessment conducted by pathologists is not optimal. Consequently, Bejnordi et al. [192] curated the CAMELYON16 dataset to focus on lymph node metastasis detection. Veeling et al. [193] proposed utilizing inherent rotation and reflection symmetries in histopathology images by employing a modified DenseNet, ultimately achieving SOTA performance.

V. DISCUSSION AND CONCLUSION

The field of medical imaging has undergone a profound evolution over the past 130 years, transitioning from Roentgen's X-ray images in 1895 to a wide range of modalities. Each modality has carved out a niche in clinical practice due to its unique strengths. X-rays remain cost-effective for bone imaging [201], CT excellent for trauma assessment and staging [202]. MRI provides superior soft-tissue contrast [203], ultrasound provides real-time radiation free visualization [204], and nuclear techniques deliver metabolic information [205], [206]. Despite these strengths, limitations persist: radiation risks in X-ray/CT [207], high costs and contraindications (e.g., metal implants) in MRI, operator dependence in ultrasound [208], and tracer accessibility in nuclear imaging [209].

In CT, ultrahigh-resolution systems that utilize the Extremely Brilliant Source (EBS) have achieved cellular-level imaging, as demonstrated by Hierarchical Phase-Contrast Tomography (HiP-CT), which scanned a human heart with 20 μ m isotropic voxels [21]. This promises to visualize critical cardiac structures, such as the His-Purkinje system, potentially revolutionizing direct pacing techniques for cardiac desynchrony [210], [211]. Likewise, improved CT resolution and new reconstruction algorithms are now enabling visualization of cardiac fibrosis [212], [213] – a capability that was previously exclusive to MRI [214] – thus offering an alternative for patients contraindicated for CRT. In MRI, higher field strengths up to 11.4 Tesla [27] and improved contrast agents [203] are enhancing soft-tissue

visualization, while ultrasound benefits from 3D and 4D imaging that provides detailed fetal and cardiac assessments. In addition, hybrid imaging systems PET/CT and PET/MRI combine anatomical and functional data to improve diagnostic accuracy and guide treatment planning. Integration of machine learning for image reconstruction is another promising direction, reducing acquisition times and improving image quality across modalities [29]. Current generation scanners that employ ML-driven algorithms minimize noise in low-dose CT scans [215], thus balancing image quality with reduced radiation exposure.

We have traced the evolution of deep learning methods for both general computer vision and medical image analysis. The results show that the trajectory of the medical imaging field primarily mirrors the progression of computer vision approaches, albeit with a delay. The ongoing architectural development of computer vision models shows the importance of new architectures – from AlexNet's [59] increased depth to ResNet's [67] residual connections and the pure transformer's attention mechanisms [82], architectures have consistently pushed models performance. While such techniques in computer vision have provided a template, the unique constraints of medical imaging demand more specific approaches to achieve clinically acceptable performance. More specifically, for medical classification tasks, the strength of deep learning lies in the extraction of discriminative features unique to each category. This is relatively straightforward for data with high interclass variance (e.g., eye fundus), where deep learning models have achieved high accuracy. However, challenges are still present for data with high inter-class similarity (e.g., microscopic images), where the subtle differences in features limit overall performance.

Segmentation tasks in medical imaging present challenges similar to those in classification. Class imbalance is a prevalent issue in both 2D and 3D segmentation, further worsened by the difficulty of accurately segmenting very small lesions or organs [99]. While modified evaluation metrics and loss functions have partially mitigated these issues [99], they primarily address region-based errors. This highlights the need to develop non-region-based metrics that capture structural, shape, and contour information, which are required for diagnostic and prognostic accuracy. A recurring theme throughout this review is the critical reliance of deep learning performance on the availability of large, annotated datasets. Supervised transfer learning offers a partial solution by leveraging pre-trained models [118] on ImageNet. Furthermore, the review highlights the potential of GANs for data augmentation [144] and self-supervised/semi-supervised learning [153] to utilize information from unlabeled data. However, as different research teams apply the same model to a task, they often demonstrate vastly different results [136]. In this regard, the development of new automated pipelines for architecture configuration, proposed by nnU-Net [118], represents a promising direction for

optimizing deep learning performance in medical image analysis.

In conclusion, this review has illustrated the substantial progress and promising potential of deep learning in medical image analysis across various clinically significant tasks, including classification and segmentation. While summarizing the significant achievements, the review also highlights the remaining technical challenges that impede the widespread clinical adoption of deep learning-based systems. Future research should prioritize addressing these challenges to facilitate the development of robust, reliable, and clinically viable deep learning solutions for medical imaging.

REFERENCES

- [1] E. Bercovich and M. C. Javitt, "Medical imaging: From roentgen to the digital revolution, and beyond," *Rambam Maimonides Med. J.*, vol. 9, no. 4, p. e0034, Oct. 2018.
- [2] J. A. Seibert, "One hundred years of medical diagnostic imaging technology," *Health Phys.*, vol. 69, no. 5, pp. 695–720, Nov. 1995.
- [3] S. Hussain, I. Mubeen, N. Ullah, S. S. U. D. Shah, B. A. Khan, M. Zahoor, R. Ullah, F. A. Khan, and M. A. Sultan, "Modern diagnostic imaging technique applications and risk factors in the medical field: A review," *BioMed Res. Int.*, vol. 2022, no. 1, Jan. 2022, Art. no. 5164970. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2022/5164970>
- [4] J. Wallyn, N. Anton, S. Akram, and T. F. Vandamme, "Biomedical imaging: Principles, technologies, clinical aspects, contrast agents, limitations and future trends in nanomedicines," *Pharmaceutical Res.*, vol. 36, no. 6, pp. 1–31, Jun. 2019.
- [5] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. V. D. Laak, B. V. Ginneken, and C. I. Sánchez, "A survey on deep learning in medical image analysis," *Med. Image Anal.*, vol. 42, pp. 60–88, Aug. 2017. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1361841517301135>
- [6] K. Suzuki, "Overview of deep learning in medical imaging," *Radiological Phys. Technol.*, vol. 10, no. 3, pp. 257–273, Sep. 2017, doi: [10.1007/s12194-017-0406-5](https://doi.org/10.1007/s12194-017-0406-5).
- [7] X. Chen, X. Wang, K. Zhang, K. Fung, T. Thai, K. N. Moore, R. S. Mannel, H. Liu, B. Zheng, and Y. Qiu, "Recent advances and clinical applications of deep learning in medical image analysis," *Med. Image Anal.*, vol. 79, Sep. 2022, Art. no. 102444. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1361841522000913>
- [8] R. Mould, "Röntgen and the discovery of X-rays," *Brit. J. Radiol.*, vol. 68, no. 815, pp. 1145–1176, 1995.
- [9] M. L. F. Nascimento, "Brief history of X-ray tube patents," *World Pat. Inf.*, vol. 37, pp. 48–53, Jun. 2014.
- [10] S. Balter, "Fluoroscopic technology from 1895 to 2019 drivers: Physics and physiology," *Med. Phys. Int.*, vol. 7, pp. 111–140, Sep. 2019.
- [11] S. D. Stranks and H. J. Snaith, "Metal-halide perovskites for photovoltaic and light-emitting devices," *Nature Nanotechnol.*, vol. 10, no. 5, pp. 391–402, May 2015.
- [12] L.-J. Xu, X. Lin, Q. He, M. Worku, and B. Ma, "Highly efficient eco-friendly X-ray scintillators based on an organic manganese halide," *Nature Commun.*, vol. 11, no. 1, p. 4329, Aug. 2020.
- [13] J. Xiong, Z.-H. Zhang, Z. Li, P. Zheng, J. Li, X. Zhang, Z. Gao, Z. Wei, G. Zheng, S.-P. Wang, and H.-C. Liu, "Perovskite single-pixel detector for dual-color metasurface imaging recognition in complex environment," *Light, Sci. Appl.*, vol. 12, no. 1, p. 286, Nov. 2023.
- [14] S. Bates, L. Beckmann, A. Thomas, and R. Waltham, *Godfrey Hounsfield: Intuitive Genius of CT*, vol. 14. London, U.K.: British Institute of Radiology London, 2012.
- [15] R. A. Schulz, J. A. Stein, and N. J. Pelc, "How CT happened: The early development of medical computed tomography," *J. Med. Imag.*, vol. 8, no. 5, Oct. 2021, Art. no. 052110.
- [16] J. Sijbers and A. Postnov, "Reduction of ring artefacts in high resolution micro-CT reconstructions," *Phys. Med. Biol.*, vol. 49, no. 14, pp. N247–N253, Jul. 2004.
- [17] W. A. Kalender, "Technical foundations of spiral computed tomography," in *Computed Tomography*, 1996, pp. 17–28.
- [18] P. Rogalla, C. Kloeters, and P. A. Hein, "CT technology overview: 64-slice and beyond," *Radiologic Clinics North Amer.*, vol. 47, no. 1, pp. 1–11, Jan. 2009.
- [19] C.-J. Lin, J.-C. Hsu, Y.-J. Lai, K.-L. Wang, J.-Y. Lee, A.-H. Li, and S.-H. Chu, "Diagnostic accuracy of dual-source CT coronary angiography in a population unselected for degree of coronary artery calcification and without heart rate modification," *Clin. Radiol.*, vol. 65, no. 2, pp. 109–117, Feb. 2010.
- [20] S. Leng, M. Bruesewitz, S. Tao, K. Rajendran, A. F. Halaweish, N. G. Campeau, J. G. Fletcher, and C. H. McCollough, "Photon-counting detector CT: System design and clinical applications of an emerging technology," *RadioGraphics*, vol. 39, no. 3, pp. 729–743, May 2019.
- [21] J. Brunet, A. C. Cook, C. L. Walsh, J. Cranley, P. Tafforeau, K. Engel, O. Arthurs, C. Berruyer, E. Burke O'Leary, A. Bellier, R. Torii, C. Werlein, D. D. Jonigk, M. Ackermann, K. Dollman, and P. D. Lee, "Multidimensional analysis of the adult human heart in health and disease using hierarchical phase-contrast tomography," *Radiology*, vol. 312, no. 1, Jul. 2024, Art. no. 232731, doi: [10.1148/radiol.232731](https://doi.org/10.1148/radiol.232731).
- [22] J. R. Mallard, "Magnetic resonance imaging—The Aberdeen perspective on developments in the early years," *Phys. Med. Biol.*, vol. 51, no. 13, pp. R45–R60, Jul. 2006.
- [23] J. T. Bushberg, J. Seibert, E. Leidholdt, and J. Boone, "Magnetic resonance basics, magnetic fields, nuclear magnetic characteristics, tissue contrast, image acquisition," *Essential Phys. Med. Imag.*, vol. 2012, pp. 402–448, Sep. 2012.
- [24] L. E. Crooks, D. A. Ortendahl, L. Kaufman, J. Hoenninger, M. Arakawa, J. Watts, C. R. Cannon, M. Brant-Zawadzki, P. L. Davis, and A. R. Margulis, "Clinical efficiency of nuclear magnetic resonance imaging," *Magn. Reson. Imag.*, vol. 2, no. 1, p. 74, Jan. 1984.
- [25] H. Fukatsu, "3T MR for clinical use: Update," *Magn. Reson. Med. Sci.*, vol. 2, no. 1, pp. 37–45, 2003.
- [26] P. J. Koopmans, M. Barth, S. Orzada, and D. G. Norris, "Multi-echo fMRI of the cortical laminae in humans at 7T," *NeuroImage*, vol. 56, no. 3, pp. 1276–1285, Jun. 2011.
- [27] Ž. Vujović, "Magnets, gradients, and RF coils of MR scanners," *Int. J. Phys. Res. Appl.*, vol. 6, no. 2, pp. 128–135, Jul. 2023.
- [28] P. M. Glover, I. Cavin, W. Qian, R. Bowtell, and P. A. Gowland, "Magnetic-field-induced vertigo: A theoretical and experimental investigation," *Bioelectromagnetics*, vol. 28, no. 5, pp. 349–361, Jul. 2007.
- [29] M. M. Bissell et al., "4D flow cardiovascular magnetic resonance consensus statement: 2023 update," *J. Cardiovascular Magn. Reson.*, vol. 25, no. 1, p. 40, Feb. 2023.
- [30] T. L. Faber, "Multiheaded rotating gamma cameras in cardiac single-photon emission computed tomographic imaging," *J. Nucl. Cardiology*, vol. 1, no. 3, pp. 292–303, May 1994.
- [31] L. Mosconi, "Glucose metabolism in normal aging and Alzheimer's disease: Methodological and physiological considerations for PET studies," *Clin. Translational Imag.*, vol. 1, no. 4, pp. 217–233, Aug. 2013.
- [32] A. A. Lammertsma, "On the origin of hybrid imaging," *J. Nucl. Med.*, vol. 61, no. 2, p. 166, 2020.
- [33] J. Ophir and N. F. Maklad, "Digital scan converters in diagnostic ultrasound imaging," *Proc. IEEE*, vol. 67, no. 4, pp. 654–664, 1979.
- [34] J. C. Somer, "The history of real time ultrasound," in *Proc. Int. Congr. Ser.*, vol. 1274, 2004, pp. 3–13. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0531513104013378>
- [35] K. Baba, K. Satoh, S. Sakamoto, T. Okai, and S. Ishii, "Development of an ultrasonic system for three-dimensional reconstruction of the fetus," *jpme*, vol. 17, no. 1, pp. 19–24, 1989, doi: [10.1515/jpme.1989.17.1.19](https://doi.org/10.1515/jpme.1989.17.1.19).
- [36] M. Murtin, D. Spitzer, and N. H. Zech, "Principles of 3D ultrasound," in *Ultrasound Imaging in Reproductive Medicine: Advances in Infertility Work-Up, Treatment, and ART*. Cham, Switzerland: Springer, 2014, pp. 17–25.
- [37] E. Betzig, G. H. Patterson, R. Sougrat, O. W. Lindwasser, S. Olenych, J. S. Bonifacio, M. W. Davidson, J. Lippincott-Schwartz, and H. F. Hess, "Imaging intracellular fluorescent proteins at nanometer resolution," *Science*, vol. 313, no. 5793, pp. 1642–1645, Sep. 2006.
- [38] O. Couture, M. Tanter, and M. Fink, "Method and device for ultrasound imaging," U.S. Patent 9 329 260, May 3, 2016.
- [39] C. Huang, W. Zhang, P. Gong, U.-W. Lok, S. Tang, T. Yin, X. Zhang, L. Zhu, M. Sang, P. Song, R. Zheng, and S. Chen, "Super-resolution ultrasound localization microscopy based on a high frame-rate clinical ultrasound scanner: An in-human feasibility study," *Phys. Med. Biol.*, vol. 66, no. 8, Apr. 2021, Art. no. 08NT01.

- [40] M. D. Brown, B. S. Generowicz, S. Dijkhuizen, S. K. E. Koekkoek, C. Strydis, J. G. Bosch, P. Arvanitis, G. Springeling, G. J. T. Leus, C. I. De Zeeuw, and P. Kruizinga, "Four-dimensional computational ultrasound imaging of brain hemodynamics," *Sci. Adv.*, vol. 10, no. 3, p. 7957, Jan. 2024.
- [41] Z. Jiang, S. E. Salcudean, and N. Navab, "Robotic ultrasound imaging: State-of-the-art and future perspectives," *Med. image Anal.*, vol. 89, Jun. 2023, Art. no. 102878.
- [42] T. Jones and D. Townsend, "History and future technical innovation in positron emission tomography," *J. Med. Imag.*, vol. 4, no. 1, Mar. 2017, Art. no. 011013.
- [43] L. Sibille, B. Chambert, S. Alonso, C. Barrau, E. D'Estanque, Y. Al Tabaa, L. Collombier, C. Demattei, P.-O. Kotzki, and V. Boudousq, "Impact of the adaptive statistical iterative reconstruction technique on radiation dose and image quality in bone SPECT/CT," *J. Nucl. Med.*, vol. 57, no. 7, pp. 1091–1095, Jul. 2016.
- [44] C. Desmonts, M. A. Bouthiba, B. Eniloric, C. Nganoa, D. Agostini, and N. Aide, "Evaluation of a new multipurpose whole-body CzT-based camera: Comparison with a dual-head anger camera and first clinical images," *EJNMMI Phys.*, vol. 7, no. 1, pp. 1–16, Dec. 2020.
- [45] J. Rong, A. Haider, T. E. Jeppesen, L. Josephson, and S. H. Liang, "Radiochemistry for positron emission tomography," *Nature Commun.*, vol. 14, no. 1, p. 3257, Jun. 2023.
- [46] S. Surti, A. Kuhn, M. E. Werner, A. E. Perkins, J. Kolthammer, and J. S. Karp, "Performance of Philips Gemini TF PET/CT scanner with special consideration for its time-of-flight imaging capabilities," *J. Nucl. Med.*, vol. 48, no. 3, pp. 471–480, 2007.
- [47] J. S. Karp, S. Surti, M. E. Daube-Witherspoon, and G. Muehlechner, "Benefit of time-of-flight in PET: Experimental and clinical results," *J. Nucl. Med.*, vol. 49, no. 3, pp. 462–470, Mar. 2008.
- [48] I. Rausch, J. Cal-González, D. Dapra, H. J. Gallowitsch, P. Lind, T. Beyer, and G. Minear, "Performance evaluation of the biograph mCT flow PET/CT system according to the NEMA NU2–2012 standard," *EJNMMI Phys.*, vol. 2, no. 1, pp. 1–17, Dec. 2015.
- [49] D. A. López-Mora, A. Flotats, F. Fuentes-Ocampo, V. Camacho, A. Fernández, A. Ruiz, J. Duch, M. Sizova, A. Domènech, M. Estorch, and I. Carrió, "Comparison of image quality and lesion detection between digital and analog PET/CT," *Eur. J. Nucl. Med. Mol. Imag.*, vol. 46, no. 6, pp. 1383–1390, Jun. 2019.
- [50] N. Aide, C. Lasnon, C. Desmonts, I. S. Armstrong, M. Walker, and D. R. McGowan, "Advances in PET/CT technology: An update," in *Seminars in Nuclear Medicine*, vol. 52. Amsterdam, The Netherlands: Elsevier, 2021, pp. 286–301.
- [51] G. Delso, S. Fürst, B. Jakoby, R. Ladebeck, C. Ganter, S. G. Nekolla, M. Schwaiger, and S. I. Ziegler, "Performance measurements of the Siemens mMR integrated whole-body PET/MR scanner," *J. Nucl. Med.*, vol. 52, no. 12, pp. 1914–1922, Dec. 2011.
- [52] O. A. Catalano, *Clinical PET/MRI*. New York, NY, USA: Academic, 2022.
- [53] Y.-D. Son, Y.-B. Kim, J.-H. Kim, J.-H. Kim, D.-H. Kwon, H. Lee, and Z.-H. Cho, "Future prospects of positron emission tomography–Magnetic resonance imaging hybrid systems and applications in psychiatric disorders," *Pharmaceuticals*, vol. 15, no. 5, p. 583, May 2022.
- [54] S. Weiss, C. A. Kulikowski, and A. Safir, "Glaucoma consultation by computer," *Comput. Biol. Med.*, vol. 8, no. 1, pp. 25–40, Jan. 1978.
- [55] J. Roehrig, T. Doi, A. Hasegawa, B. Hunt, J. Marshall, H. Romsdahl, A. Schneider, R. Sharbaugh, and W. Zhang, *Clinical Results With R2 Imagechecker System*. Cham, Switzerland: Springer, 1998, pp. 395–400.
- [56] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel, "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, Dec. 1989.
- [57] Y. C. Wu, K. Doi, and M. L. Giger, "Detection of lung nodules in digital chest radiographs using artificial neural networks: A pilot study," *J. Digit. Imag.*, vol. 8, no. 2, pp. 88–94, May 1995.
- [58] K. Doi, M. L. Giger, R. M. Nishikawa, and R. A. Schmidt, "Computer aided diagnosis of breast cancer on mammograms," *Breast cancer*, vol. 4, no. 4, pp. 228–233, 1997.
- [59] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 60, 2017, pp. 84–90.
- [60] H. R. Roth, C. T. Lee, H.-C. Shin, A. Seff, L. Kim, J. Yao, L. Lu, and R. M. Summers, "Anatomy-specific classification of medical images using deep convolutional nets," in *Proc. IEEE 12th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2015, pp. 101–104.
- [61] Y. Bar, I. Diamant, L. Wolf, and H. Greenspan, "Deep learning with non-medical training used for chest pathology identification," *Proc. SPIE*, vol. 9414, pp. 215–221, Mar. 2015.
- [62] Simonyan, "Very deep convolutional networks for large-scale image recognition," in *Proc. 3rd Int. Conf. Learn. Represent.*, 2015, pp. 1–14.
- [63] A. Akselrod-Ballin, L. Karlinsky, S. Alpert, S. Hasoul, R. Ben-Ari, and E. Barkan, "A region based convolutional network for tumor detection and classification in breast mammography," in *Deep Learning and Data Labeling for Medical Applications*. Cham, Switzerland: Springer, 2016, pp. 197–205.
- [64] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.
- [65] J. Chang, J. Yu, T. Han, H.-j. Chang, and E. Park, "A method for classifying medical images using transfer learning: A pilot study on histopathology of breast cancer," in *Proc. IEEE 19th Int. Conf. e-Health Neww. Appl. Services (Healthcom)*, Oct. 2017, pp. 1–4.
- [66] N. Alsharman and I. Jawarneh, "GoogleNet CNN neural network towards chest CT-coronavirus medical image classification," *J. Comput. Sci.*, vol. 16, no. 5, pp. 620–625, May 2020.
- [67] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [68] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5987–5995.
- [69] Y. Ishikawa, K. Washiya, K. Aoki, and H. Nagahashi, "Brain tumor classification of microscopy images using deep residual learning," *Proc. SPIE*, vol. 10013, pp. 222–231, Dec. 2016.
- [70] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.
- [71] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng, "CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning," 2017, *arXiv:1711.05225*.
- [72] S. Otálora, M. Atzori, V. Andrearczyk, and H. Müller, "Image magnification regression using DenseNet for exploiting histopathology open access content," in *Proc. Int. Workshop Ophthalmic Med. Image Anal.* Cham, Switzerland: Springer, 2018, pp. 148–155.
- [73] S.-C.-B. Lo, H.-P. Chan, J.-S. Lin, H. Li, M. T. Freedman, and S. K. Mun, "Artificial convolution neural network for medical image pattern recognition," *Neural Netw.*, vol. 8, nos. 7–8, pp. 1201–1214, 1995. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0893608095000615>
- [74] B. Sahiner, H.-P. Chan, N. Petrick, D. Wei, M. A. Helvie, D. D. Adler, and M. M. Goodsitt, "Classification of mass and normal breast tissue: A convolution neural network classifier with spatial domain and texture images," *IEEE Trans. Med. Imag.*, vol. 15, no. 5, pp. 598–610, Jun. 1996.
- [75] Shallu and R. Mehra, "Breast cancer histology images classification: Training from scratch or transfer learning?" *ICT Exp.*, vol. 4, no. 4, pp. 247–254, Dec. 2018. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2405959518304934>
- [76] M. Limam and F. Precioso, "Atrial fibrillation detection and ECG classification based on convolutional recurrent neural network," in *Proc. Comput. Cardiol. (CinC)*, Sep. 2017, pp. 1–4.
- [77] I. Koné and L. Boulmane, "Hierarchical ResNeXt models for breast cancer histology image classification," in *Image Analysis and Recognition*, A. Campilho, F. Karray, and B. ter Haar Romeny, Eds., Cham, Switzerland: Springer, 2018, pp. 796–803.
- [78] C. P. Bridge, M. Rosenthal, B. Wright, G. Kotecha, F. Fintelmann, F. Troschel, N. Miskin, K. Desai, W. Wrobel, A. Babic, N. Khalaf, L. Brais, M. Welch, C. Zellers, N. Tenenholtz, M. Michalski, B. Wolpin, and K. Andriole, "Fully-automated analysis of body composition from CT in cancer patients using convolutional neural networks," in *Proc. Int. Workshop Comput.-Assist. Robotic Endoscopy*. Cham, Switzerland: Springer, 2018, pp. 204–213.

- [79] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
- [80] N. Gessert, M. Nielsen, M. Shaikh, R. Werner, and A. Schlaefer, "Skin lesion classification using ensembles of multi-resolution EfficientNets with meta data," *MethodsX*, vol. 7, Mar. 2020, Art. no. 100864. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2215016120300832>
- [81] J. Wang, L. Yang, Z. Huo, W. He, and J. Luo, "Multi-label classification of fundus images with EfficientNet," *IEEE Access*, vol. 8, pp. 212499–212508, 2020.
- [82] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
- [83] S. Perera, S. Adhikari, and A. Yilmaz, "Pocformer: A lightweight transformer architecture for detection of covid-19 using point of care ultrasound," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 195–199.
- [84] B. Gheflati and H. Rivaz, "Vision transformers for classification of breast ultrasound images," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2022, pp. 480–483.
- [85] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11976–11986.
- [86] G. Tian, Z. Wang, C. Wang, J. Chen, G. Liu, H. Xu, Y. Lu, Z. Han, Y. Zhao, Z. Li, X. Luo, and L. Peng, "A deep ensemble learning-based automated detection of COVID-19 using lung CT images and vision transformer and ConvNeXt," *Frontiers Microbiol.*, vol. 13, Nov. 2022, Art. no. 1024104.
- [87] S. Liu, T. Cai, X. Tang, Y. Zhang, and C. Wang, "COVID-19 diagnosis via chest X-ray image classification based on multiscale class residual attention," *Comput. Biol. Med.*, vol. 149, Oct. 2022, Art. no. 106065. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0010482522007764>
- [88] S. Woo, S. Debnath, R. Hu, X. Chen, Z. Liu, I. S. Kweon, and S. Xie, "ConvNeXt v2: Co-designing and scaling ConvNets with masked autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 16133–16142.
- [89] B. Ozdemir and I. Pacal, "An innovative deep learning framework for skin cancer detection employing ConvNeXtV2 and focal self-attention mechanisms," *Results Eng.*, vol. 25, Mar. 2025, Art. no. 103692. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590123024019352>
- [90] W. Yu, P. Zhou, S. Yan, and X. Wang, "InceptionNeXt: When inception meets ConvNeXt," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5672–5683.
- [91] B. Ozdemir, E. Aslan, and I. Pacal, "Attention enhanced InceptionNeXt-based hybrid deep learning model for lung cancer detection," *IEEE Access*, vol. 13, pp. 27050–27069, 2025.
- [92] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.
- [93] Y. Haruna, S. Qin, A. H. Adama Chukkol, A. A. Yusuf, I. Bello, and A. Lawan, "Exploring the synergies of hybrid convolutional neural network and vision transformer architectures for computer vision: A survey," *Eng. Appl. Artif. Intell.*, vol. 144, Mar. 2025, Art. no. 110057. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0952197625000570>
- [94] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
- [95] X. Zhou, T. Ito, R. Takayama, S. Wang, T. Hara, and H. Fujita, "Three-dimensional CT image segmentation by combining 2D fully convolutional network with 3D majority voting," in *Deep Learning and Data Labeling for Medical Applications*. Cham, Switzerland: Springer, 2016, pp. 111–120.
- [96] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.
- [97] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning dense volumetric segmentation from sparse annotation," in *Proc. 19th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Athens, Greece. Cham, Switzerland: Springer, Oct. 2016, pp. 424–432.
- [98] F. Wang, R. Jiang, L. Zheng, C. Meng, and B. B. Biswal, "3D U-Net based brain tumor segmentation and survival days prediction," in *Proc. Int. MICCAI Brainlesion Workshop*, 2020, pp. 131–141.
- [99] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
- [100] G. Litjens et al., "Evaluation of prostate segmentation algorithms for MRI: The PROMISE12 challenge," *Med. Image Anal.*, vol. 18, no. 2, pp. 359–373, Feb. 2014.
- [101] M. Z. Alom, C. Yakopcic, M. Hasan, T. M. Taha, and V. K. Asari, "Recurrent residual U-Net for medical image segmentation," *J. Med. Imag.*, vol. 6, no. 1, Mar. 2019, Art. no. 014006.
- [102] O. Oktay, J. Schlemper, L. Le Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, B. Glocker, and D. Rueckert, "Attention U-Net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*.
- [103] L. Yu, S. Wang, X. Li, C. Fu, and P. Heng, "Uncertainty-aware self-ensembling model for semi-supervised 3D left atrium segmentation," in *Proc. 22nd Int. Conf. Med. Image Comput. Comput. Assist. Intervent.*, Shenzhen, China. Cham, Switzerland: Springer, Oct. 2019, pp. 605–613.
- [104] (2018). *Atriaseg 2018 Challenge*. [Online]. Available: <https://www.cardiacatlas.org/atriseg2018-challenge/>
- [105] A. Mehrtash, W. M. Wells, C. M. Tempany, P. Abolmaesumi, and T. Kapur, "Confidence calibration and predictive uncertainty estimation for deep medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 12, pp. 3868–3878, Dec. 2020.
- [106] J. Chen, J. Mei, X. Li, Y. Lu, Q. Yu, Q. Wei, X. Luo, Y. Xie, E. Adeli, Y. Wang, M. P. Lungren, S. Zhang, L. Xing, L. Lü, A. Yuille, and Y. Zhou, "TransUNet: Rethinking the U-Net architecture design for medical image segmentation through the lens of transformers," *Proc. Med. Image Anal.*, vol. 97, Oct. 2024, Art. no. 103280. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1361841524002056>
- [107] Synapse. (2025). *Synapse Wiki Page for Synapse: Syn3193805*. [Online]. Available: <https://www.synapse.org/Synapse:syn3193805/wiki/217789>
- [108] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, "Swin-unet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2021, pp. 205–218.
- [109] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
- [110] (2025). *ACDC Challenge*. [Online]. Available: <https://www.creatis.insa-lyon.fr/Challenge/acdc/>
- [111] Y. Zhang, H. Liu, and Q. Hu, "TransFuse: Fusing transformers and CNNs for medical image segmentation," in *Proc. 24th Int. Conf. Med. Image Comput. Comput. Assist. Intervent.*, Strasbourg, France. Cham, Switzerland: Springer, Sep. 2021, pp. 14–24.
- [112] D. Jha, P. H. Smedsrud, M. A. Riegler, P. Halvorsen, T. D. Lange, D. Johansen, and H. D. Johansen, "Kvasir-SEG: A segmented polyp dataset," in *Proc. 26th Int. Conf. MultiMedia Modeling*. Cham, Germany: Springer, 2019, pp. 451–462.
- [113] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in *Proc. Int. MICCAI Brainlesion Workshop*, 2022, pp. 272–284.
- [114] (2021). *Brain Tumor Segmentation Challenge*. [Online]. Available: <http://braintumorsegmentation.org/>
- [115] Q. Xu, Z. Ma, H. Na, and W. Duan, "DCSAU-net: A deeper and more compact split-attention U-net for medical image segmentation," *Comput. Biol. Med.*, vol. 154, Mar. 2023, Art. no. 106626.
- [116] J. Bernal, F. J. Sánchez, G. Fernández-Esparrach, D. Gil, C. Rodríguez, and F. Vilariño, "WM-DOVA maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians," *Computerized Med. Imag. Graph.*, vol. 43, pp. 99–111, Jul. 2015.

- [117] N. C. F. Codella, D. Gutman, M. E. Celebi, B. Helba, M. A. Marchetti, S. W. Dusza, A. Kalloo, K. Liopyris, N. Mishra, H. Kittler, and A. Halpern, "Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (ISBI), hosted by the international skin imaging collaboration (ISIC)," in *Proc. IEEE 15th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2018, pp. 168–172.
- [118] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-net: A self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, no. 2, pp. 203–211, Feb. 2021.
- [119] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, and Y. Zhou, "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.
- [120] L. Lan, P. Cai, L. Jiang, X. Liu, Y. Li, and Y. Zhang, "BRAU-Net++: U-shaped hybrid CNN-Transformer network for medical image segmentation," 2024, *arXiv:2401.00722*.
- [121] S. Zheng, X. Ye, C. Yang, L. Yu, W. Li, X. Gao, and Y. Zhao, "Asymmetric adaptive heterogeneous network for multi-modality medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 44, no. 4, pp. 1836–1852, Apr. 2025.
- [122] P. Cai, L. Jiang, Y. Li, and L. Lan, "Pubic symphysis-fetal head segmentation using pure transformer with bi-level routing attention," 2023, *arXiv:2310.00289*.
- [123] T. Amit, T. Shaharabany, E. Nachmani, and L. Wolf, "SegDiff: Image segmentation with diffusion probabilistic models," 2021, *arXiv:2112.00390*.
- [124] J. Wu, H. Fang, Y. Zhang, Y. Yang, Y. Xu, X. Haoyi, L. Huiying, and X. Yanwu, "MedSegDiff: Medical image segmentation with diffusion probabilistic model," in *Proc. Med. Imag. Deep Learn.*, Jul. 2022, pp. 1623–1639. [Online]. Available: <https://proceedings.mlr.press/v227/wu24a.html>
- [125] J. Wu, W. Ji, H. Fu, M. Xu, Y. Jin, and Y. Xu, "MedSegDiff-v2: Diffusion-based medical image segmentation with transformer," in *Proc. 28th AAAI Conf. Artif. Intell., 36th Conf. Innov. Appl. Artif. Intell., 14th Symp. Educ. Adv. Artif. Intell.*, vol. 38, 2024, pp. 6030–6038, doi: [10.1609/aaai.v38i6.28418](https://doi.org/10.1609/aaai.v38i6.28418).
- [126] Z. Xing, L. Wan, H. Fu, G. Yang, Y. Yang, L. Yu, B. Lei, and L. Zhu, "Diff-UNet: A diffusion embedded network for robust 3D medical image segmentation," *Med. Image Anal.*, vol. 105, Oct. 2025, Art. no. 103654. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1361841525002014>
- [127] A. M. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W. Lo, P. Dollár, and R. Girshick, "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 3992–4003.
- [128] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer, "SAM 2: Segment anything in images and videos," 2024, *arXiv:2408.00714*.
- [129] J. Cheng, J. Ye, Z. Deng, J. Chen, T. Li, H. Wang, Y. Su, Z. Huang, J. Chen, L. Jiang, H. Sun, J. He, S. Zhang, M. Zhu, and Y. Qiao, "SAM-Med2D," 2023, *arXiv:2308.16184*.
- [130] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, "Segment anything in medical images," *Nature Commun.*, vol. 15, no. 1, pp. 1–16, Jan. 2024.
- [131] A. Archi, L. Freckmann, and C. Pape, "MedicoSAM: Robust improvement of SAM for medical imaging," 2025, *arXiv:2501.11734*.
- [132] S. Gatidis, T. Hepp, M. Früh, C. L. Fougère, K. Nikolaou, C. Pfannenberger, B. Schölkopf, T. Küstner, C. Cyran, and D. Rubin, "A whole-body FDG-PET/CT dataset with manually annotated tumor lesions," *Scientific Data*, vol. 9, no. 1, p. 601, Oct. 2022.
- [133] M. Hadlich, Z. Marinov, and R. Stiefelhagen, "AutoPET challenge 2023: Sliding window-based optimization of U-Net," 2023, *arXiv:2309.12114*.
- [134] Y. Peng, J. Kim, D. Feng, and L. Bi, "Automatic tumor segmentation via false positive reduction network for whole-body multi-modal PET/CT images," 2022, *arXiv:2209.07705*.
- [135] J. Ye, H. Wang, Z. Huang, Z. Deng, Y. Su, C. Tu, Q. Wu, Y. Yang, M. Wei, J. Niu, and J. He, "Exploring vanilla U-Net for lesion segmentation from whole-body FDG-PET/CT scans," 2022, *arXiv:2210.07490*.
- [136] J. Wasserthal, H.-C. Breit, M. T. Meyer, M. Pradella, D. Hinck, A. W. Sauter, T. Heye, D. T. Boll, J. Cyriac, S. Yang, M. Bach, and M. Segeroth, "TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images," *Radiol. Artif. Intell.*, vol. 5, no. 5, Sep. 2023, Art. no. 230024.
- [137] Z. Huang, H. Wang, Z. Deng, J. Ye, Y. Su, H. Sun, J. He, Y. Gu, L. Gu, S. Zhang, and Y. Qiao, "STU-Net: Scalable and transferable medical image segmentation models empowered by large-scale supervised pre-training," 2023, *arXiv:2304.06716*.
- [138] T. Akinci D'Antonoli, L. K. Berger, A. K. Indrakanti, N. Vishwanathan, J. Weiss, M. Jung, Z. Berkarda, A. Rau, M. Reisert, T. Küstner, A. Walter, E. M. Merkle, D. T. Boll, H.-C. Breit, A. P. Nicoli, M. Segeroth, J. Cyriac, S. Yang, and J. Wasserthal, "TotalSegmentator MRI: Robust sequence-independent segmentation of multiple anatomic structures in MRI," *Radiology*, vol. 314, no. 2, pp. 1–14, Feb. 2025.
- [139] M. Antonelli et al., "The medical segmentation decathlon," *Nature Commun.*, vol. 13, no. 1, p. 4128, 2022.
- [140] B. Rister, D. Yi, K. Shivakumar, T. Nobashi, and D. L. Rubin, "CT-ORG, a new dataset for multiple organ segmentation in computed tomography," *Scientific Data*, vol. 7, no. 1, p. 381, Nov. 2020.
- [141] R. K. Debnath, M. A. Rahman, S. Azam, Y. Zhang, and M. Jonkman, "FSS-ULivR: A clinically-inspired few-shot segmentation framework for liver imaging using unified representations and attention mechanisms," *J. Cancer Res. Clin. Oncol.*, vol. 151, no. 7, pp. 1–23, Jul. 2025.
- [142] U. Baid et al., "The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification," 2021, *arXiv:2107.02314*.
- [143] S. Roy, G. Koehler, C. Ulrich, M. Baumgartner, J. Petersen, F. Isensee, P. F. Jaeger, and K. Maier-Hein, "MedNeXt: transformer-driven scaling of ConvNets for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2023, pp. 405–415.
- [144] A. Ferreira, N. Solak, J. Li, P. Dammann, J. Kleesiek, V. Alves, and J. Egger, "How we won BraTS 2023 adult glioma challenge? Just faking it! Enhanced synthetic data augmentation and model ensemble for brain tumour segmentation," 2024, *arXiv:2402.17317*.
- [145] F. Bolelli et al., "Segmenting the inferior alveolar canal in CBCT volumes: The ToothFairy challenge," *IEEE Trans. Med. Imag.*, vol. 44, no. 4, pp. 1890–1906, Apr. 2025.
- [146] L. Lumetti, V. Pipoli, F. Bolelli, E. Ficarra, and C. Grana, "Enhancing patch-based learning for the segmentation of the mandibular canal," *IEEE Access*, vol. 12, pp. 79014–79024, 2024.
- [147] F. Bolelli, K. Marchesini, N. Van Nistelrooij, L. Lumetti, V. Pipoli, E. Ficarra, S. Vinayahalingam, and C. Grana, "Segmenting maxillofacial structures in CBCT volumes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 5238–5248.
- [148] H. Gong, G. Chen, R. Wang, X. Xie, M. Mao, Y. Yu, F. Chen, and G. Li, "Multi-task learning for thyroid nodule segmentation with thyroid region prior," in *Proc. IEEE 18th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2021, pp. 257–261.
- [149] H. Gong, J. Chen, G. Chen, H. Li, G. Li, and F. Chen, "Thyroid region prior guided attention for ultrasound segmentation of thyroid nodules," *Comput. Biol. Med.*, vol. 155, Mar. 2023, Art. no. 106389.
- [150] P. Dong, R. Zhang, J. Li, C. Liu, W. Liu, J. Hu, Y. Yang, and X. Li, "An ultrasound image segmentation method for thyroid nodules based on dual-path attention mechanism-enhanced UNet++," *BMC Med. Imag.*, vol. 24, no. 1, p. 341, Dec. 2024.
- [151] Z. Huang, *Mmwhs: Medical Multi-Modal Whole Slide Image Segmentation*. Accessed: Nov. 21, 2025. [Online]. Available: <https://zmiclab.github.io/zxh0/mmwhs/>
- [152] C. Chen, Q. Dou, H. Chen, J. Qin, and P. Heng, "Synergistic image and feature adaptation: Towards cross-modality domain adaptation for medical image segmentation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 865–872.
- [153] H. Basak, R. Bhattacharya, R. Hussain, and A. Chatterjee, "An exceedingly simple consistency regularization method for semi-supervised medical image segmentation," in *Proc. IEEE 19th Int. Symp. Biomed. Imag. (ISBI)*, Mar. 2022, pp. 1–4.
- [154] J. Yang, X. Ding, Z. Zheng, X. Xu, and X. Li, "GraphEcho: Graph-driven unsupervised domain adaptation for echocardiogram video segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 11844–11853.
- [155] B. Pu, L. Wang, J. Yang, G. He, X. Dong, S. Li, Y. Tan, M. Chen, Z. Jin, K. Li, and X. Li, "M3-UDA: A new benchmark for unsupervised domain adaptive fetal cardiac structure detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 11621–11630.
- [156] X. Zhuang, "Multivariate mixture model for myocardial segmentation combining multi-source images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 12, pp. 2933–2946, Dec. 2019.

- [157] J. Qiu, L. Li, S. Wang, K. Zhang, Y. Chen, S. Yang, and X. Zhuang, "MyoPS-Net: Myocardial pathology segmentation with flexible combination of multi-sequence CMR images," *Med. Image Anal.*, vol. 84, Feb. 2023, Art. no. 102694. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S136184152200322X>
- [158] K.-N. Wang, X. Yang, J. Miao, L. Li, J. Yao, P. Zhou, W. Xue, G.-Q. Zhou, X. Zhuang, and D. Ni, "AWSNet: An auto-weighted supervision attention network for myocardial scar and edema segmentation in multi-sequence cardiac magnetic resonance images," *Med. Image Anal.*, vol. 77, Apr. 2022, Art. no. 102362.
- [159] D. F. Pace, H. T. M. Contreras, J. Romanowicz, S. Ghelani, I. Rahaman, Y. Zhang, P. Gao, M. I. Jubair, T. Yeh, P. Golland, T. Geva, S. Ghelani, A. J. Powell, and M. H. Moghari, "HVSMMR-2.0: A 3D cardiovascular MR dataset for whole-heart segmentation in congenital heart disease," *Sci. Data*, vol. 11, no. 1, p. 721, Jul. 2024.
- [160] Y. Zhu, H. Li, B. Cao, K. Huang, and J. Liu, "A novel hybrid layer-based encoder-decoder framework for 3D segmentation in congenital heart disease," *Sci. Rep.*, vol. 15, no. 1, p. 11891, Apr. 2025.
- [161] N. Ghamsarian, Y. El-Shabrawi, S. Nasirihaghighi, D. Putzgruber-Adamitsch, M. Zinkernagel, S. Wolf, K. Schoeffmann, and R. Sznitman, "Cataract-1K dataset for deep-learning-assisted analysis of cataract surgery videos," *Scientific Data*, vol. 11, no. 1, p. 373, Apr. 2024.
- [162] H. B. Li et al., "The brain tumor segmentation (BraTS) challenge 2023: Brain MR image synthesis for tumor segmentation (BraSyn)," 2024, *arXiv:2305.09011v6*.
- [163] M. Correia de Verdiere et al., "The 2024 brain tumor segmentation (BraTS) challenge: Glioma segmentation on post-treatment MRI," 2024, *arXiv:2405.18368*.
- [164] J. Ma, F. Li, and B. Wang, "U-Mamba: Enhancing long-range dependency for biomedical image segmentation," 2024, *arXiv:2401.04722*.
- [165] L. Li et al., "MyoPS: A benchmark of myocardial pathology segmentation combining three-sequence cardiac magnetic resonance images," *Med. Image Anal.*, vol. 87, Jul. 2023, Art. no. 102808.
- [166] T. Karkera, C. Adak, S. Chattopadhyay, and M. Saqib, "Detecting severity of diabetic retinopathy from fundus images: A transformer network-based review," *Neurocomputing*, vol. 597, Sep. 2024, Art. no. 127991.
- [167] H. Shakibania, S. Raoufi, B. Pourafkham, H. Khotanlou, and M. Mansoorizadeh, "Dual branch deep learning network for detection and stage grading of diabetic retinopathy," *Biomed. Signal Process. Control*, vol. 93, Jul. 2024, Art. no. 106168.
- [168] (2019). *Aptos 2019 Blindness Detection*. [Online]. Available: <https://www.kaggle.com/competitions/aptos2019-blindness-detection>
- [169] S. Camalan, M. K. K. Niazi, A. C. Moberly, T. Teknos, G. Essig, C. Elmaraghy, N. Taj-Schaal, and M. N. Gurcan, "OtoMatch: Content-based eardrum image retrieval using deep learning," *PLoS ONE*, vol. 15, no. 5, May 2020, Art. no. e0232776.
- [170] M. A. Rahman, M. A. Raiaan, S. Azam, A. Karim, J. Beissbarth, and A. Leach, "WeCKD: Weakly-supervised chained distillation network for efficient multimodal medical imaging," 2025, *arXiv:2510.14668*.
- [171] C. Matek, S. Krappe, C. Münzenmayer, T. Haferlach, and C. Marr, "Highly accurate differentiation of bone marrow cell morphologies using deep neural networks on a large image data set," *Blood*, vol. 138, no. 20, pp. 1917–1927, Nov. 2021, doi: [10.1182/blood.2020010568](https://doi.org/10.1182/blood.2020010568).
- [172] J. Jantzen, J. Norup, G. Dounias, and B. Bjerregaard, "Pap-smear benchmark data for pattern classification," *Nature Inspired Smart Inf. Syst.*, vol. 2005, pp. 1–9, Oct. 2005.
- [173] S. Sornapudi, G. T. Brown, Z. Xue, R. Long, L. Allen, and S. Antani, "Comparing deep learning models for multi-cell classification in liquid-based cervical cytology image," in *AMIA Annu. Symp. Proc.*, 2019, p. 820.
- [174] H. Basak, R. Kundu, S. Chakraborty, and N. Das, "Cervical cytology classification using PCA and GWO enhanced deep features selection," *Social Netw. Comput. Sci.*, vol. 2, no. 5, p. 369, Sep. 2021.
- [175] M. E. Plissiti, P. Dimitrakopoulos, G. Sfikas, C. Nikou, O. Krikoni, and A. Charchanti, "Sipakmed: A new dataset for feature and image based classification of normal and pathological cervical cells in pap smear images," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3144–3148.
- [176] A. Manna, R. Kundu, D. Kaplun, A. Sinitca, and R. Sarkar, "A fuzzy rank-based ensemble of CNN models for classification of cervical cytology," *Sci. Rep.*, vol. 11, no. 1, p. 14538, Jul. 2021.
- [177] J. Silva-Rodríguez, A. Colomer, M. A. Sales, R. Molina, and V. Naranjo, "Going deeper through the Gleason scoring scale: An automatic end-to-end system for histology prostate grading and cribriform pattern detection," *Comput. Methods Programs Biomed.*, vol. 195, Oct. 2020, Art. no. 105637.
- [178] H. Bian, Z. Shao, Y. Chen, Y. Wang, H. Wang, J. Zhang, and Y. Zhang, "Multiple instance learning with mixed supervision in Gleason grading," in *Proc. Int. Conf. Med. Image Comput. Computer-Assisted Intervent*, 2022, pp. 204–213.
- [179] W. Bulten et al., "Artificial intelligence for diagnosis and Gleason grading of prostate cancer: The PANDA challenge," *Nature Med.*, vol. 28, no. 1, pp. 154–163, 2022.
- [180] R. J. Chen, T. Ding, M. Y. Lu, D. F. K. Williamson, G. Jaume, A. H. Song, B. Chen, A. Zhang, D. Shao, M. Shaban, M. Williams, L. Oldenburg, L. A. Weishaupt, J. J. Wang, A. Vaidya, L. P. Le, G. Gerber, S. Sahai, W. Williams, and F. Mahmood, "Towards a general-purpose foundation model for computational pathology," *Nature Med.*, vol. 30, no. 3, pp. 850–862, Mar. 2024.
- [181] (2025). *PPMI Information*. [Online]. Available: <https://www.ppmi-info.org>
- [182] A. Kurmi, S. Biswas, S. Sen, A. Sinitca, D. Kaplun, and R. Sarkar, "An ensemble of CNN models for Parkinson's disease detection using DaTscan images," *Diagnostics*, vol. 12, no. 5, p. 1173, May 2022.
- [183] X. Yang, X. He, J. Zhao, Y. Zhang, S. Zhang, and P. Xie, "COVID-CT-dataset: A CT scan dataset about COVID-19," 2020, *arXiv:2003.13865*.
- [184] S. Tabik, A. Gómez-Ríos, J. L. Martín-Rodríguez, I. Sevillano-García, M. Rey-Area, D. Charre, E. Guirado, J. L. Suárez, J. Luengo, M. A. Valero-González, P. García-Villanova, E. Olmedo-Sánchez, and F. Herrera, "COVIDGR dataset and COVID-SDNet methodology for predicting COVID-19 based on chest X-ray images," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 12, pp. 3595–3605, Dec. 2020.
- [185] P. Lakhani et al., "The 2021 SIIM-FISABIO-RSNA machine learning COVID-19 challenge: Annotation and standard exam classification of COVID-19 chest radiographs," *J. Digit. Imag.*, vol. 36, no. 1, pp. 365–372, Sep. 2022.
- [186] A. Elhanashi, S. Saponara, and Q. Zheng, "Classification and localization of multi-type abnormalities on chest X-rays images," *IEEE Access*, vol. 11, pp. 83264–83277, 2023.
- [187] W. Hu, C. Li, X. Li, M. M. Rahaman, J. Ma, Y. Zhang, H. Chen, W. Liu, C. Sun, Y. Yao, H. Sun, and M. Grzegorzec, "GasHisSDB: A new gastric histopathology image dataset for computer aided diagnosis of gastric cancer," *Comput. Biol. Med.*, vol. 142, Mar. 2022, Art. no. 105207.
- [188] M. P. Yong, Y. C. Hum, K. W. Lai, Y. L. Lee, C.-H. Goh, W.-S. Yap, and Y. K. Tee, "Histopathological gastric cancer detection on GasHisSDB dataset using deep ensemble learning," *Diagnostics*, vol. 13, no. 10, p. 1793, May 2023.
- [189] K. Kriegsmann, F. Lobers, C. Zgorzelski, J. Kriegsmann, C. Janßen, R. R. Meliß, T. Muley, U. Sack, G. Steinbusch, and M. Kriegsmann, "Deep learning for the detection of anatomical tissue structures and neoplasms of the skin on scanned histopathological tissue sections," *Frontiers Oncol.*, vol. 12, Nov. 2022, Art. no. 1022967.
- [190] N. R. Kurtansky et al., "The SLICE-3D dataset: 400,000 skin lesion image crops extracted from 3D TBP for skin cancer detection," *Sci. Data*, vol. 11, no. 1, p. 884, Aug. 2024.
- [191] A. El Mrabet, M. Benaly, I. Alihamidi, B. Kouach, L. Hlou, and R. E. Gouri, "Enhancing early detection of skin cancer in clinical practice with hybrid deep learning models," *Eng., Technol. Appl. Sci. Res.*, vol. 15, no. 2, pp. 20927–20933, Apr. 2025.
- [192] B. E. Bejnordi et al., "Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer," *Jama*, vol. 318, no. 22, p. 2199, 2017.
- [193] B. S. Veeling, J. Linmans, J. Winkens, T. Cohen, and M. Welling, "Rotation equivariant CNNs for digital pathology," in *Proc. 21st Int. Conf. Med. Image Comput. Comput. Assist. Intervent.*, Granada, Spain. Cham, Switzerland: Springer, Sep. 2018, pp. 210–218.
- [194] Y. Liu, K. Gadepalli, M. Norouzi, G. E. Dahl, T. Kohlberger, A. Boyko, S. Venugopalan, A. Timofeev, P. Q. Nelson, G. S. Corrado, J. D. Hipp, L. Peng, and M. C. Stumpe, "Detecting cancer metastases on gigapixel pathology images," 2017, *arXiv:1703.02442*.
- [195] R. R. A. Bourne et al., "Prevalence and causes of vision loss in high-income countries and in eastern and central Europe in 2015: Magnitude, temporal trends and projections," *Brit. J. Ophthalmol.*, vol. 102, no. 5, pp. 575–585, May 2018.

- [196] Y. Zhang, Y. Wei, Q. Wu, P. Zhao, S. Niu, J. Huang, and M. Tan, "Collaborative unsupervised domain adaptation for medical image diagnosis," *IEEE Trans. Image Process.*, vol. 29, pp. 7834–7844, 2020.
- [197] F.-H. Wang et al., "The Chinese society of clinical oncology (CSCO): Clinical guidelines for the diagnosis and treatment of gastric cancer," *Cancer Commun.*, vol. 39, no. 1, pp. 1–31, Dec. 2019.
- [198] H. Sung, J. Ferlay, R. L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, and F. Bray, "Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA, A Cancer J. Clinicians*, vol. 71, no. 3, pp. 209–249, May 2021.
- [199] C. Garbe et al., "Skin cancers are the most frequent cancers in fair-skinned populations, but we can prevent them," *Eur. J. Cancer*, vol. 204, Jun. 2024, Art. no. 114074.
- [200] M. A. Rahman, N. M. Fahad, M. A. K. Raiaan, M. Jonkman, F. De Boer, and S. Azam, "Advancing skin cancer detection integrating a novel unsupervised classification and enhanced imaging techniques," *CAAI Trans. Intell. Technol.*, vol. 10, no. 2, pp. 474–493, Apr. 2025.
- [201] X. Ou, X. Chen, X. Xu, L. Xie, X. Chen, Z. Hong, H. Bai, X. Liu, Q. Chen, L. Li, and H. Yang, "Recent development in X-ray imaging technology: Future and challenges," *Research*, vol. 2021, pp. 1–15, Jan. 2021. [Online]. Available: <https://spj.science.org/doi/abs/10.34133/2021/9892152>
- [202] S. Dighe, S. Purkayastha, I. Swift, P. P. Tekkis, A. Darzi, R. A'Hern, and G. Brown, "Diagnostic precision of CT in local staging of colon cancers: A meta-analysis," *Clin. Radiol.*, vol. 65, no. 9, pp. 708–719, Sep. 2010.
- [203] S. M. Smith, M. Jenkinson, M. W. Woolrich, C. F. Beckmann, T. E. J. Behrens, H. Johansen-Berg, P. R. Bannister, M. De Luca, I. Drobniak, D. E. Flitney, R. K. Niazy, J. Saunders, J. Vickers, Y. Zhang, N. De Stefano, J. M. Brady, and P. M. Matthews, "Advances in functional and structural MR image analysis and implementation as FSL," *NeuroImage*, vol. 23, pp. S208–S219, Jan. 2004. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1053811904003933>
- [204] D. L. Miller, N. B. Smith, M. R. Bailey, G. J. Czarnota, K. Hynynen, and I. R. S. Makin, "Overview of therapeutic ultrasound applications and safety considerations," *J. Ultrasound Med.*, vol. 31, no. 4, pp. 623–634, Apr. 2012.
- [205] E. Franquet and H. Park, "Molecular imaging in oncology: Common PET/CT radiopharmaceuticals and applications," *Eur. J. Radiol. Open*, vol. 9, Oct. 2022, Art. no. 100455.
- [206] L. Zhu, K. Ploessl, and H. F. Kung, "PET/SPECT imaging agents for neurodegenerative diseases," *Chem. Soc. Rev.*, vol. 43, no. 19, pp. 6683–6691, 2014.
- [207] D. P. Frush, M. J. Callahan, B. D. Coley, H. R. Nadel, and R. P. Guillerman, "Comparison of the different imaging modalities used to image pediatric oncology patients: A COG diagnostic imaging committee/SPR oncology committee white paper," *Pediatric Blood Cancer*, vol. 70, no. S4, p. 30298, Jun. 2023.
- [208] A. L. Aagesen and M. Melek, "Choosing the right diagnostic imaging modality in musculoskeletal diagnosis," *Primary Care, Clinics Office Pract.*, vol. 40, no. 4, pp. 849–861, Dec. 2013.
- [209] F. Pisaneschi and N. T. Viola, "Development and validation of a PET/SPECT radiopharmaceutical in oncology," *Mol. Imag. Biol.*, vol. 24, no. 1, pp. 1–7, Feb. 2022.
- [210] A. Zweerink, S. Zubarev, E. Bakelants, D. Potyagaylo, C. Stettler, M. Chmelevsky, E. D. Lozeron, A.-L. Hachulla, J.-P. Vallée, and H. Burri, "His-optimized cardiac resynchronization therapy with ventricular fusion pacing for electrical resynchronization in heart failure," *JACC, Clin. Electrophysiol.*, vol. 7, no. 7, pp. 881–892, Jul. 2021.
- [211] H. Burri, M. Jastrzebski, Ó. Cano, K. Curila, J. de Pooter, W. Huang, C. Israel, J. Joza, J. Romero, K. Vernooy, P. Vijayaraman, Z. Whinnett, and F. Zanon, "EHRA clinical consensus statement on conduction system pacing implantation: Executive summary. Endorsed by the asia-pacific heart rhythm society (APHRS), Canadian Heart Rhythm Society (CHRS) and Latin-American Heart Rhythm Society (LAHRS)," *Europace*, vol. 25, no. 4, pp. 1237–1248, Apr. 2023.
- [212] N. Y. Kim, D. J. Im, Y. J. Hong, B. W. Choi, S.-M. Kang, J.-C. Youn, and H.-J. Lee, "Feasibility of the threshold-based quantification of myocardial fibrosis on cardiac CT as a prognostic marker in nonischemic dilated cardiomyopathy," *Korean J. Radiol.*, vol. 25, no. 6, p. 540, 2024.
- [213] P. R. Scully, K. P. Patel, E. Klotz, J. B. Augusto, G. D. Thornton, B. Saberwal, U. Haberland, S. Kennon, M. Ozkor, M. Mullen, G. Lloyd, A. Kelion, L. J. Menezes, P. N. Hawkins, J. C. Moon, F. Pugliese, and T. A. Treibel, "Myocardial fibrosis quantified by cardiac CT predicts outcome in severe aortic stenosis after transcatheter intervention," *JACC, Cardiovascular Imag.*, vol. 15, no. 3, pp. 542–544, Mar. 2022.
- [214] A. Bazhutina, S. Zubarev, and M. Budanova, "Accuracy of cardiac scar and fibrosis geometry in ventricular models affects electrical dyssynchrony biomarkers and ML prediction score of CRT response," in *Proc. Ural-Siberian Conf. Comput. Technol. Cognit. Sci., Genomics Biomed. (CSGB)*, Jul. 2022, pp. 280–283.
- [215] M. Beister, D. Kolditz, and W. A. Kalender, "Iterative reconstruction methods in X-ray CT," *Phys. Medica*, vol. 28, no. 2, pp. 94–108, Apr. 2012.



ALEKSANDR LONCHAKOV received the B.S. degree in mechatronics and robotics and the M.S. degree in computer technologies in robotics from Ufa State Aviation Technical University, Ufa, Russia, in 2018 and 2024, respectively. He is currently pursuing the Ph.D. degree with Saint Petersburg Electrotechnical University "LETI," Saint Petersburg, Russia. His research interest includes the application of machine learning to medical image analysis.



ALEKSANDR SINITCA (Member, IEEE) received the B.S. and M.S. degrees in computer science from the Faculty of Computer Science and Informatics, Saint Petersburg Electrotechnical University "LETI," Saint Petersburg, Russia, in 2015 and 2017, respectively. He is currently a Senior Researcher with the Department of Radio Engineering Systems, Saint Petersburg Electrotechnical University "LETI." He is the author of more than 50 papers in journals and conference proceedings. His research interests include medical image processing, nonlinear dynamics, and industrial automation and informatics.



DMITRII KAPLUN (Member, IEEE) received the Ph.D. degree in digital signal processing from Saint Petersburg Electrotechnical University "LETI," Saint Petersburg, Russia, in 2009. He was an Associate Professor with Saint Petersburg Electrotechnical University "LETI," in 2015, where he is currently a Lead Researcher. He regularly takes part in different collaborative projects related to signal processing, machine learning, and embedded computing. The most substantial results are in the fields of digital signal and image processing, embedded systems, computer vision, and machine learning. He is the author of more than 100 papers in journals and conference proceedings. His current research and academic work are related to digital signal and image processing, radio monitoring and hydroacoustic monitoring applications, embedded and reconfigurable systems, computer vision, and machine learning.

...